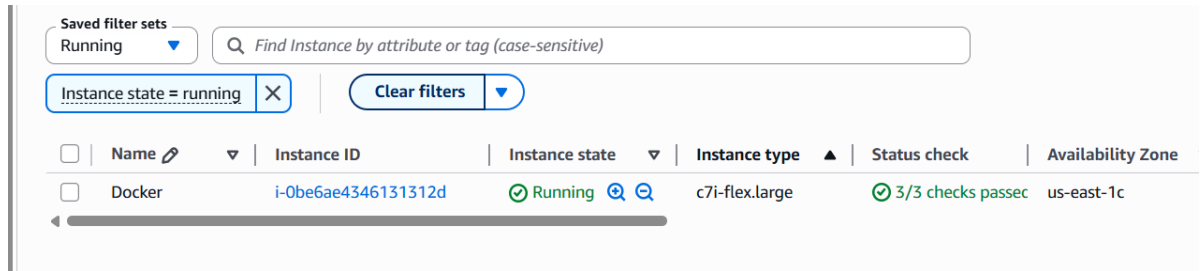


JENKINS -04

- 1) Setup Jenkins CICD pipeline using freestyle job using Docker containers using below code.

Source Code: <https://github.com/betawins/hiring-app.git>

- Created new EC2 instance as docker



- Connected the ec2 and install the docker in it

```
[ec2-user@ip-172-31-27-142 ~]$ sudo su -
[root@ip-172-31-27-142 ~]# sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository 366 kB/s | 38 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-27-142 ~]# sudo yum install docker -y
Last metadata expiration check: 0:00:12 ago on Wed May 20 06:11:13 2026.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
docker x86_64 25.0.14-1.amzn2023.0.5 amazonlinux 46 M
Installing dependencies:
container-selinux noarch 4:2.245.0-1.amzn2023 amazonlinux 58 k
```

- Docker version

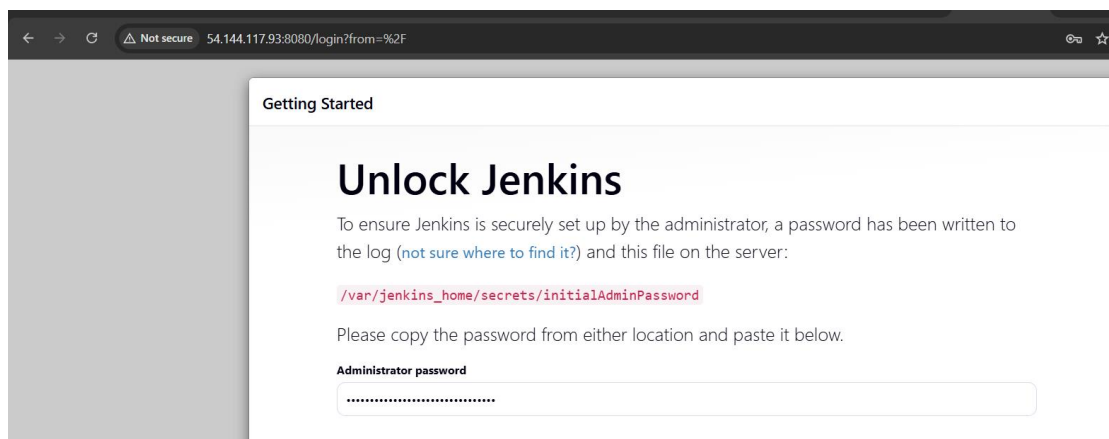
```
[root@ip-172-31-27-142 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-27-142 ~]# docker network create cicd-network
1ebc16d2b5a7929cad21a19573f9e2585e1513b1de0e5c2e30871642109eef99
[root@ip-172-31-27-142 ~]# docker run -d \
--name jenkins \
```

- In that installed Jenkins by using below command

```
docker run -d \
--name jenkins \
--network cicd-network \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
jenkins/jenkins: lts
```

```
[root@ip-172-31-27-142 ~]# docker run -d \
--name jenkins \
--network cicd-network \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
307f8152a55e: Pull complete
cebc1e77ceba: Pull complete
```

- Public id of ec2 of docker and with port no: 8080 open in browser

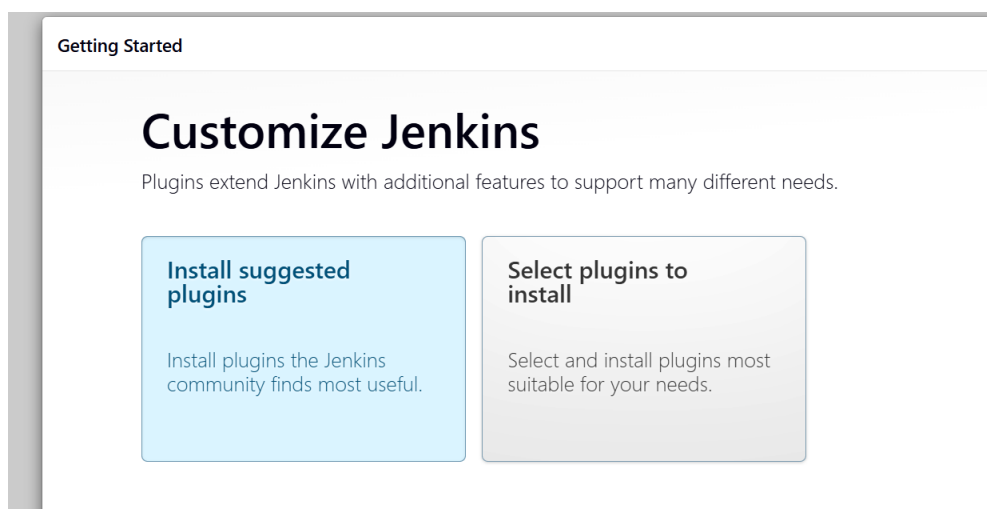


- Generated password in server by using below command

docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword

```
2a7d946e27d86c4c5cf5148aa0224c83c5eb8feb642f7485cbe89d9ab45f6948
[root@ip-172-31-27-142 ~]# docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
fff6587a900542c8acaa7be4ad279536
```

- Installed suggested plugins



- Then create user name and details after that finish and next

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build

- Jenkins is Ready and open it

Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

[Start using Jenkins](#)

Stages:

- **Git Clone**

- open jenkins and create a new item by selecting name and type Freestyle

New Item

Enter an item name

Hiring-App1

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

- Select Source Code Management in that git and paste the given Source code URL

Configure

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None
 ☒ **Git** ?

Repositories ?

Repository URL ?

`https://github.com/betawins/hiring-app.git`

Credentials ?

- none -

- Build Now and check the console we get success

Jenkins / Hiring-App1 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Console

```

Started by user Kathi Arunsai
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Hiring-App1
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/hiring-app.git
> git init /var/jenkins_home/workspace/Hiring-App1 # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git
> git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953ea1f9669784f9d6a (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9405b107fdffdf68e69fe953ea1f9669784f9d6a # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
Finished: SUCCESS
  
```

• SonarQube Integration

- Create a New Ec2 instance and connected it

☐	Docker	i-0be6ae4346131312d	Running		c7i-flex.large	3/3 checks passed	us-east-1c	ec2-54-144-117-93.co...
☐	Sonarqube-ser...	i-065b53781e28d1cfc	Running		c7i-flex.large	Initializing	us-east-1c	ec2-13-222-188-207.co...

- Install java 1.8* version and check version

```
on most systems).
[ec2-user@ip-172-31-30-70 ~]$ sudo su -
[root@ip-172-31-30-70 ~]# yum install java-1.8*
Amazon Linux 2023 Kernel Livepatch repository 336 kB/s | 38 kB 00:00
Dependencies resolved.
=====
Package                        Arch Version                               Repository Size
=====
Installing:
```

- Install MySQL community with wget URL and repo URL

```
Complete!
[root@ip-172-31-30-70 ~]# wget https://dev.mysql.com/get/mysql57-community-release-e17-11.noarch.rpm
--2026-05-14 16:04:42-- https://dev.mysql.com/get/mysql57-community-release-e17-11.noarch.rpm
Resolving dev.mysql.com (dev.mysql.com)... 23.13.181.57, 2600:1408:c400:188c::2e31, 2600:1408:c400:1881::2e31
Connecting to dev.mysql.com (dev.mysql.com)|23.13.181.57|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://repo.mysql.com//mysql57-community-release-e17-11.noarch.rpm [following]
--2026-05-14 16:04:42-- https://repo.mysql.com//mysql57-community-release-e17-11.noarch.rpm
Resolving repo.mysql.com (repo.mysql.com)... 23.33.203.94, 2600:1408:c400:168d::1d68, 2600:1408:c400:1692::1d68
Connecting to repo.mysql.com (repo.mysql.com)|23.33.203.94|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 25680 (25K) [application/x-redhat-package-manager]
Saving to: 'mysql57-community-release-e17-11.noarch.rpm'

mysql57-community-rel 100%[=====>] 25.08K --.-KB/s in 0.001s
```

- Install the MYSQL community release

```
[root@ip-172-31-30-70 ~]# yum localinstall mysql57-community-release-e17-11.noarch.rpm
Last metadata expiration check: 0:01:07 ago on Thu May 14 16:03:46 2026.
Dependencies resolved.
=====
Package                        Architecture Version                               Repository Size
=====
Installing:
mysql57-community-release      noarch          e17-11                                @commandline 25 k
Transaction Summary
=====
Install 1 Package
```

- Start MYSQL and check MYSQL port no:3306 correct or not

```
Installed:
libxcrypt-compat-4.4.33-7.amzn2023.x86_64
mysql-community-client-5.7.44-1.el7.x86_64
mysql-community-common-5.7.44-1.el7.x86_64
mysql-community-libs-5.7.44-1.el7.x86_64
mysql-community-server-5.7.44-1.el7.x86_64
ncurses-compat-libs-6.6-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-30-70 ~]# systemctl start mysqld.service
[root@ip-172-31-30-70 ~]# systemctl start mysqld
[root@ip-172-31-30-70 ~]# systemctl enable mysqld
[root@ip-172-31-30-70 ~]# netstat -na | grep 3306
tcp6      0      0 :::3306          :::*              LISTEN
[root@ip-172-31-30-70 ~]# grep 'temporary' /var/log/mysqld.log
2026-05-14T16:05:52.100249Z 1 [Note] A temporary password is generated for root@localhost: #fpPSf1l6,%7
2026-05-14T16:05:54.401093Z 0 [Note] InnoDB: Creating shared tablespace for temporary tables
```

- Login into MYSQL and generate password of local host and login with-it

```
[root@ip-172-31-30-70 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.44

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
```

- Alter the password or change the password new one and login with-it

```
mysql> ALTER USER 'root'@'localhost' IDENTIFIED BY 'Arunasai@437'
-> AC
mysql> ALTER USER 'root'@'localhost' IDENTIFIED BY 'Arunasai@437';
Query OK, 0 rows affected (0.00 sec)

mysql> flush privileges;
Query OK, 0 rows affected (0.01 sec)

mysql> exit
Bye
[root@ip-172-31-30-70 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 3
Server version: 5.7.44 MySQL Community Server (GPL)
```

- Now install SonarQube server

```
Bye
[root@ip-172-31-30-70 ~]# cd /opt
[root@ip-172-31-30-70 opt]# wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-6.6.zip
--2026-05-14 16:09:46-- https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-6.6.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.167.37.10, 3.167.37.88, 3.167.37.95, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|3.167.37.10|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 164350442 (157M) [application/zip]
Saving to: 'sonarqube-6.6.zip'

sonarqube-6.6.zip    100%[=====>] 156.74M  247MB/s  in 0.6s
```

- Then unzip the SonarQube and rename it

```
creating: sonarqube-6.6/elasticsearch/plugins/
[root@ip-172-31-30-70 opt]# mv /opt/sonarqube-6.6 /opt/sonar
[root@ip-172-31-30-70 opt]# ls
aws sonar sonarqube-6.6.zip
[root@ip-172-31-30-70 opt]# mysql -u root -p
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
[root@ip-172-31-30-70 opt]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 5
```

- Then Login to MYSQL and create table and give some database details related to sonar

```
mysql> CREATE DATABASE sonar CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.00 sec)

mysql> CREATE USER sonar@localhost IDENTIFIED BY 'Sonar@123';
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE USER sonar@%' IDENTIFIED BY 'Sonar@123';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL ON sonar.* TO sonar@localhost;
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL ON sonar.* TO sonar@'%';
Query OK, 0 rows affected (0.00 sec)

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sonar |
| sys |
+-----+
5 rows in set (0.00 sec)
```

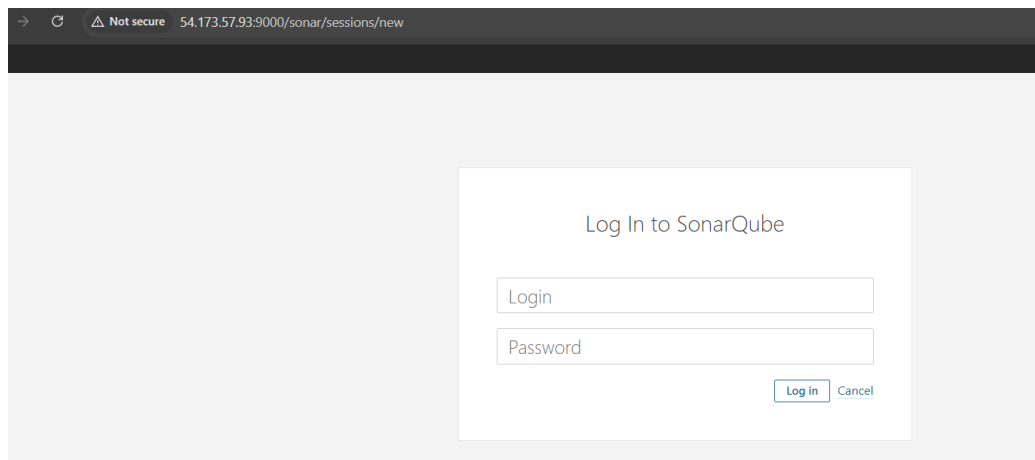
- Then add user new and given password to it

```
mysql> SELECT User FROM mysql.user;
+-----+
| User |
+-----+
| sonar |
+-----+
[root@ip-172-31-30-70 opt]# useradd Arunsai
[root@ip-172-31-30-70 opt]# passwd Arunsai
Changing password for user Arunsai.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

- Log logout it and change owner name given permissions and start SonarQube

```
Arunsai@ip-172-31-30-70 logs]$ sudo su -
[sudo] password for Arunsai:
last login: Thu May 14 16:03:08 UTC 2026 on pts/1
root@ip-172-31-30-70 ~]# chown -R Arunsai:Arunsai /opt/sonar
root@ip-172-31-30-70 ~]# chmod -R 775 /opt/sonar
root@ip-172-31-30-70 ~]# chmod -R 775 /opt/sonar
root@ip-172-31-30-70 ~]# ls -ld /opt/sonar
lrwxrwxr-x. 11 Arunsai Arunsai 141 Oct 20 2017 /opt/sonar
root@ip-172-31-30-70 ~]# sudo su - Arunsai
last login: Thu May 14 16:40:07 UTC 2026 on pts/2
Arunsai@ip-172-31-30-70 ~]$ cd /opt/sonar/bin/linux-x86-64
Arunsai@ip-172-31-30-70 linux-x86-64]$ ./sonar.sh start
starting SonarQube...
started SonarQube.
Arunsai@ip-172-31-30-70 linux-x86-64]$ ./sonar.sh status
```

- Open SonarQube in browser with SonarQube public and port no: 9000/sonar
- Login with username as admin and password as admin



- Now in docker Ec2 under Jenkins (root) installed Sonar-scanner

```
fff6587a900542c8acaa7be4ad279536
[root@ip-172-31-27-142 ~]# cd /opt
[root@ip-172-31-27-142 opt]# curl -fL -o sonar-scanner.zip \
https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cl
i-5.0.1.3006-linux.zip
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 46019k 100 46019k  0     0 333.0M    0  --:--:-- --:--:-- --:--:-- 335.3M
[root@ip-172-31-27-142 opt]# ls
aws containerd sonar-scanner.zip
[root@ip-172-31-27-142 opt]# cd ..
```

- Unzip it and Rename sonar scanner

```
m
[root@ip-172-31-27-142 /]# cd /opt
unzip sonar-scanner.zip
Archive:  sonar-scanner.zip
  creating: sonar-scanner-5.0.1.3006-linux/
  creating: sonar-scanner-5.0.1.3006-linux/jre/
  creating: sonar-scanner-5.0.1.3006-linux/jre/lib/
  creating: sonar-scanner-5.0.1.3006-linux/jre/lib/security/
```

- Install vim in it

```
bash: apt-get: command not found
[root@ip-172-31-27-142 /]# yum install vim -y
Last metadata expiration check: 0:14:36 ago on Wed May 20 06:11:13 2026.
Package vim-enhanced-2:9.2.240-1.amzn2023.0.3.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-27-142 /]# vim --version
VIM - Vi IMproved 9.2 (2026 Feb 14, compiled Apr 28 2026 00:00:00)
Included patches: 1-240, 272, 276, 357
Modified by <amazon-linux-engine@amazon.com>
Compiled by <amazon-linux-engine@amazon.com>
Huge version without GUI.  Features included (+) or not (-):
```

- Open sonar scanner properties and edit https URL and save it

```
root@ip-172-31-27-142:/opt/sonar-scanner/conf
#Configure here general information about the environment, such as SonarQube server connection details for example
#No information about specific project should appear here

#----- Default SonarQube server
sonar.host.url=http://13.222.188.207:9000/sonar

#----- Default source code encoding
#sonar.sourceEncoding=UTF-8

~
~
~
```

- In Jenkins Install Sonar Scanner Plugin

Install

Name ↓

Released

Health

☒

SonarQube Scanner 2.18.2

[External Site/Tool Integrations](#) [Build Reports](#)

This plugin allows an easy integration of [SonarQube](#), the open source platform for Continuous Inspection of code quality.

5 mo 9 days ago

84

☐

Sonar Gerrit 428.v5c962d271b_a_5

[External Site/Tool Integrations](#)

4 mo 13 days ago

96

- In SonarQube login it and in security generate token

A

Administrator

[Profile](#) [Security](#) [Notifications](#) [Projects](#)

Tokens

If you want to enforce security by not providing credentials of a real SonarQube user to run your code scan or to invoke web services, you can provide a User Token as a replacement of the user login. This will increase the security of your installation by not letting your analysis user's password going through your network.

Name	Created	
Sonarqube-api	May 14, 2026	Revoke
Sonarqube-token	May 20, 2026	Revoke

Generate New Token:

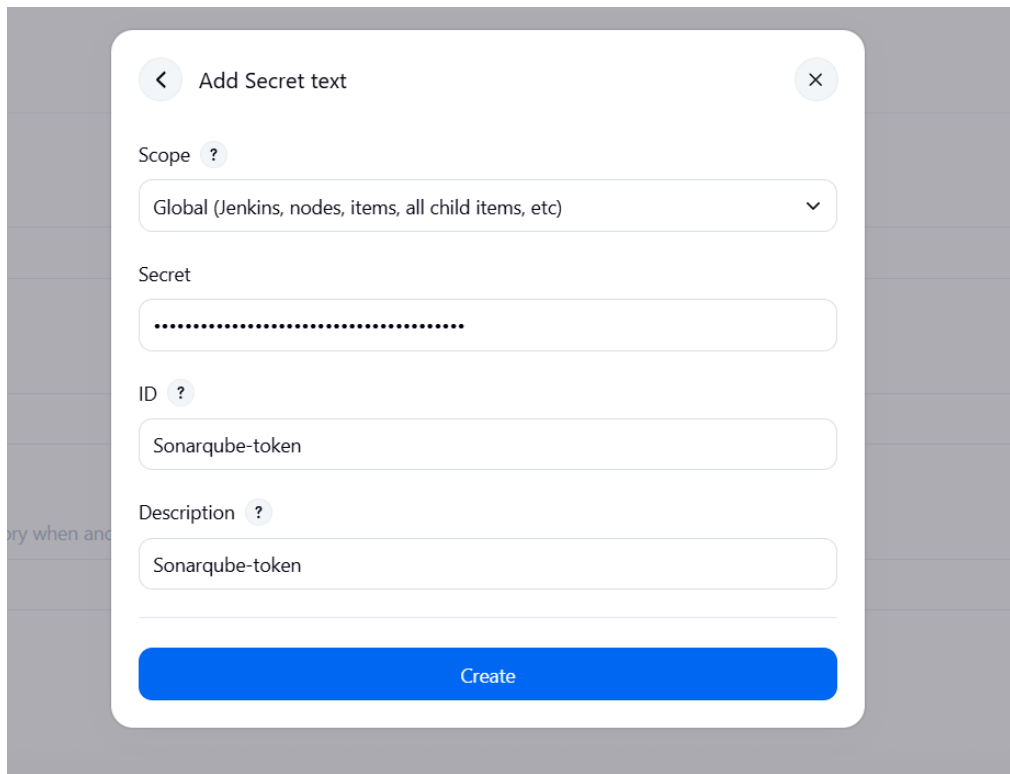
[Generate](#)

New token "Sonarqube-token" has been created. Make sure you copy it now, you won't be able to see it again!

[Copy](#)

64a0ae63fa4bfdcfc19eff8ea9186928103c2b87

- In Jenkins open credentials and create for SonarQube token



< Add Secret text ×

Scope ?
Global (Jenkins, nodes, items, all child items, etc) ▼

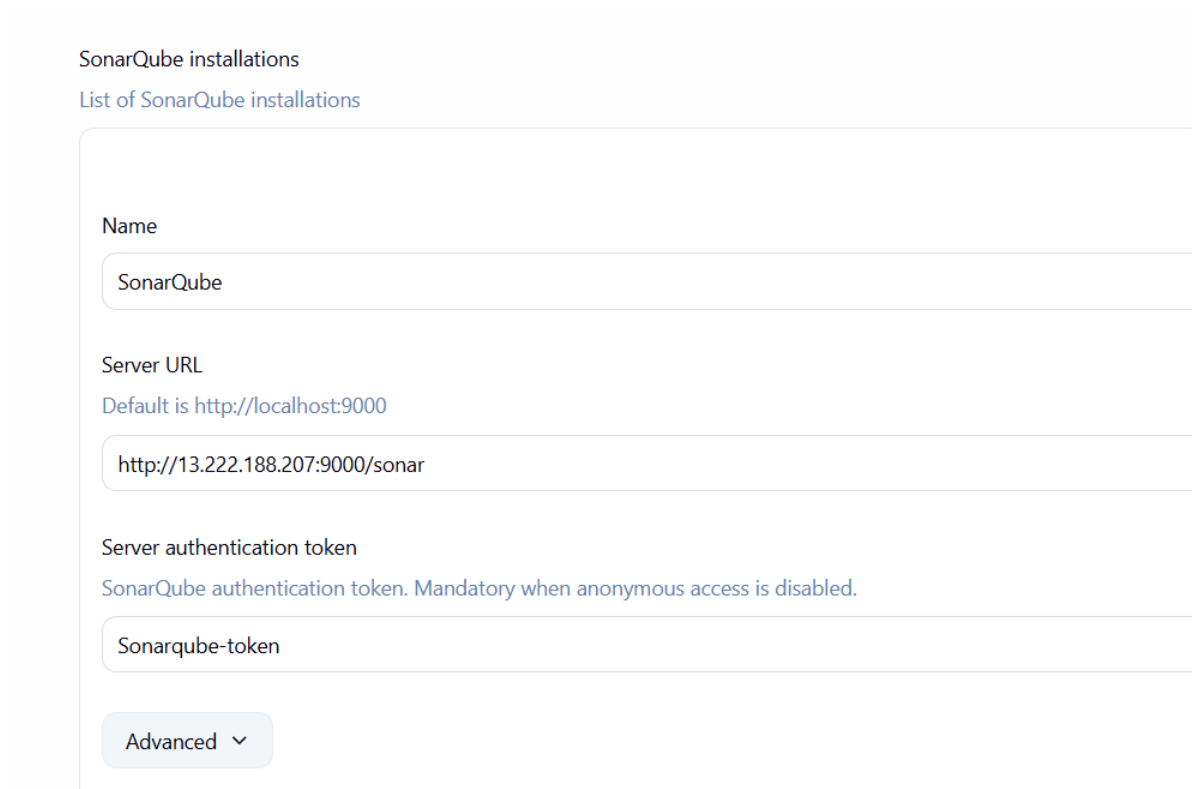
Secret
.....

ID ?
Sonarqube-token

Description ?
Sonarqube-token

Create

- In Jenkins open system and add SonarQube installation, URL and token save it



SonarQube installations

[List of SonarQube installations](#)

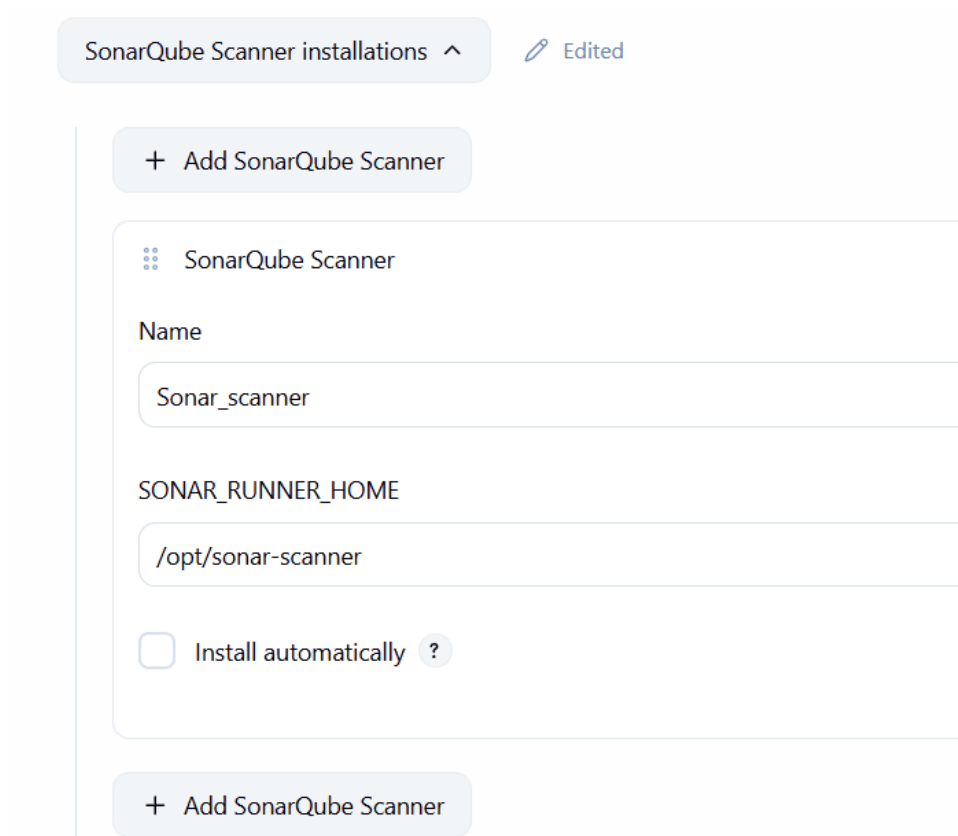
Name
SonarQube

Server URL
Default is [http://localhost:9000](#)
http://13.222.188.207:9000/sonar

Server authentication token
SonarQube authentication token. Mandatory when anonymous access is disabled.
Sonarqube-token

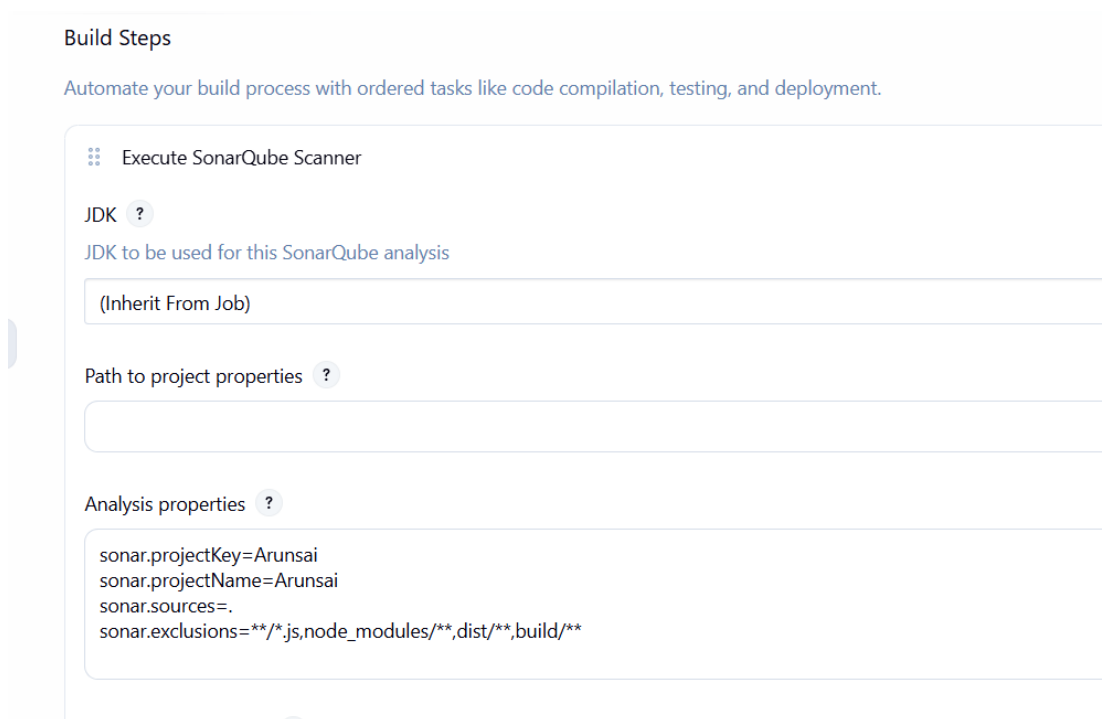
Advanced ▼

- Open Jenkins in that tools add sonar scanner details



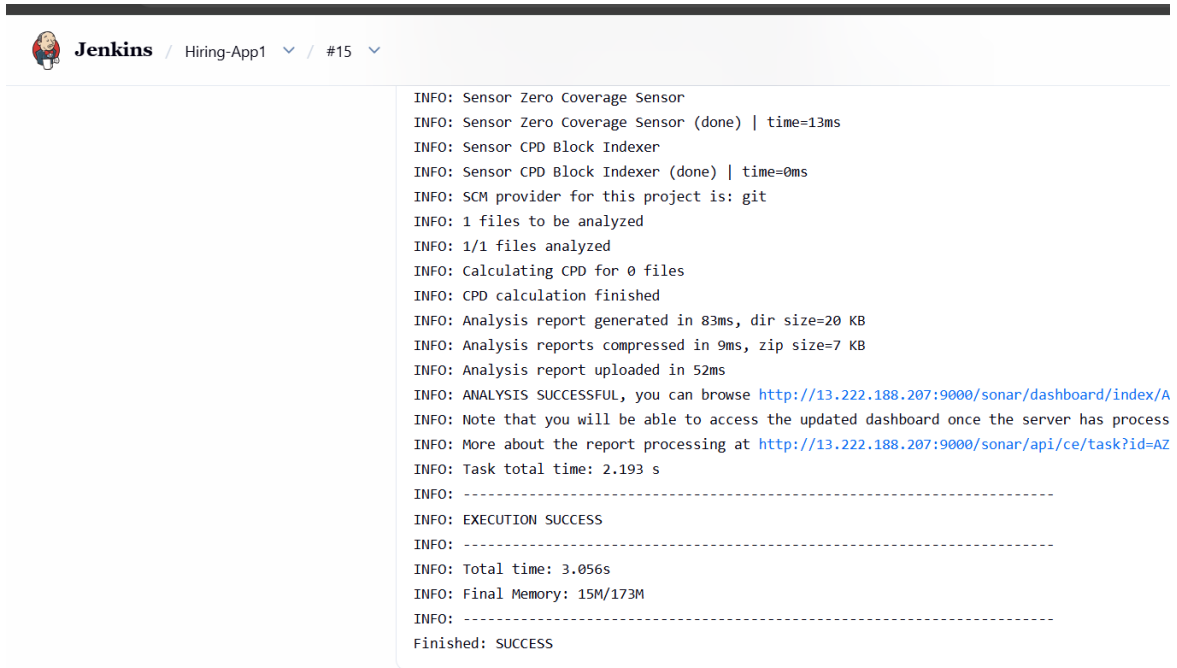
The screenshot shows the 'SonarQube Scanner installations' configuration page in Jenkins. At the top, there is a header 'SonarQube Scanner installations ^' and an 'Edited' status with a pencil icon. Below this is a button '+ Add SonarQube Scanner'. A list of installations is shown, with one entry 'SonarQube Scanner' expanded. This entry has a 'Name' field with the value 'Sonar_scanner' and a 'SONAR_RUNNER_HOME' field with the value '/opt/sonar-scanner'. There is an unchecked checkbox for 'Install automatically' with a help icon. At the bottom, there is another '+ Add SonarQube Scanner' button.

- Open create job and in configure build steps add sonar scanner details and save it



The screenshot shows the 'Build Steps' configuration page in Jenkins. The title is 'Build Steps' with a subtitle 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' Below this is a list of build steps, with one entry 'Execute SonarQube Scanner' expanded. This entry has a 'JDK' field with a help icon and a dropdown menu showing '(Inherit From Job)'. There is a 'Path to project properties' field with a help icon and an empty text input. Below this is an 'Analysis properties' field with a help icon and a text area containing the following properties:
sonar.projectKey=Arunsai
sonar.projectName=Arunsai
sonar.sources=.
sonar.exclusions=**/*.js,node_modules/**,dist/**,build/**

- Now Build Now and check the Console of it



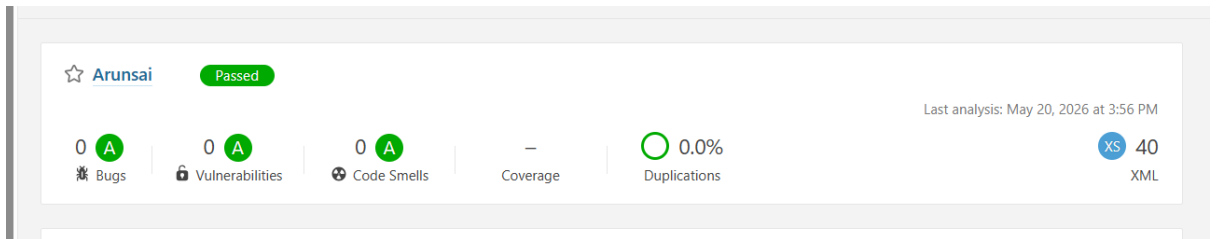
The image shows the Jenkins console output for a task named 'Hiring-App1'. The output displays the progress of a SonarQube analysis, including sensor execution, file analysis, and report generation. The task concludes with a 'SUCCESS' status.

```

INFO: Sensor Zero Coverage Sensor
INFO: Sensor Zero Coverage Sensor (done) | time=13ms
INFO: Sensor CPD Block Indexer
INFO: Sensor CPD Block Indexer (done) | time=0ms
INFO: SCM provider for this project is: git
INFO: 1 files to be analyzed
INFO: 1/1 files analyzed
INFO: Calculating CPD for 0 files
INFO: CPD calculation finished
INFO: Analysis report generated in 83ms, dir size=20 KB
INFO: Analysis reports compressed in 9ms, zip size=7 KB
INFO: Analysis report uploaded in 52ms
INFO: ANALYSIS SUCCESSFUL, you can browse http://13.222.188.207:9000/sonar/dashboard/index/A
INFO: Note that you will be able to access the updated dashboard once the server has process
INFO: More about the report processing at http://13.222.188.207:9000/sonar/api/ce/task?id=AZ
INFO: Task total time: 2.193 s
INFO: -----
INFO: EXECUTION SUCCESS
INFO: -----
INFO: Total time: 3.056s
INFO: Final Memory: 15M/173M
INFO: -----
Finished: SUCCESS

```

- Now check in SonarQube URL in projects passed successful



The image shows the SonarQube project dashboard for a project named 'Arunsai'. The project status is 'Passed'. The dashboard displays various quality metrics: 0 Bugs, 0 Vulnerabilities, 0 Code Smells, 0% Coverage, and 0.0% Duplications. The last analysis was performed on May 20, 2026 at 3:56 PM. A '40 XML' badge is also visible.

• Maven Compilation

- Now Install Maven in dockers- >jenkins ->sonar-scanner -> Maven under jenkins by using command **apt install maven -y**

```

[root@ip-172-31-27-142 ~]# docker exec -it -u root jenkins bash
root@2a7d946e27d8:/# apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
20 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  maven

Installing dependencies:
  alsa-topology-conf  libapparmor1  libcommons-lang3-java  libgu
  alsa-ucm-conf       libasound2-data  libcommons-logging-java  libha
  ca-certificates-java  libasound2t64  libcommons-parent-java  libht
  dbus                libatinject-jsr330-api-java  libcups2t64  libht
  dbus-bin            libatomic1     libdbus-1-3  libja
  dbus-daemon         libavahi-client3  liberror-prone-java  libin

```

- Check maven version and copy the maven home location



```
Setting up default-jre-headless (2:1.21-76) ...
Setting up maven (3.9.9-1) ...
update-alternatives: using /usr/share/maven/bin/mvn to provide /usr/bin/mvn (mvn) in auto mode
root@2a7d946e27d8:/# mvn -version
Apache Maven 3.9.9
Maven home: /usr/share/maven
Java version: 21.0.11, vendor: Eclipse Adoptium, runtime: /opt/java/openjdk
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.170-213.321.amzn2023.x86_64", arch: "amd64", family: "unix"
root@2a7d946e27d8:/# client_loop: send disconnect: Connection reset by peer
```

- Installed Maven Integration Plugins from plugin in jenkins



- In Jenkins open tools and add maven name and path

Maven installations

Maven installations ^

Edited

+ Add Maven

Maven

Name

Maven

MAVEN_HOME

/usr/share/maven

☐ Install automatically ?

- Open Job and configure build steps Invoke maven targets and add details

Invoke top-level Maven targets ?

Maven Version

Maven

Goals

clean package

Advanced ▾

- Now save it and build now and check in console success.

Jenkins

Hiring-App1 ▾ / #16 ▾

Downloaded[90m from @[0mcentral@[90m: https://repo.maven.apache.org/maven2/@[0morg/tukaani/xz/1.8/xz-1.8.jar@[90m (109 kB at 654 kB/s)@[0m
Downloaded[90m from @[0mcentral@[90m: https://repo.maven.apache.org/maven2/@[0morg/apache/maven/shared/maven-filtering/3.1.1/maven-filtering-3.1.1.jar@[90m (51 kB at 306 kB/s)@[0m
Progress (2): 263 kB | 4.1/11 kB
Progress (2): 263 kB | 7.7/11 kB
Progress (2): 263 kB | 11 kB

Downloaded[90m from @[0mcentral@[90m: https://repo.maven.apache.org/maven2/@[0morg/codehaus/plexus/plexus-utils/3.3.0/plexus-utils-3.3.0.jar@[90m (263 kB at 1.5 MB/s)@[0m
Downloaded[90m from @[0mcentral@[90m: https://repo.maven.apache.org/maven2/@[0morg/apache/maven/shared/maven-mapping/3.0.0/maven-mapping-3.0.0.jar@[90m (11 kB at 56 kB/s)@[0m
[0m[1;34mINFO@[0m] Packaging webapp
[0m[1;34mINFO@[0m] Assembling webapp [hiring] in [/var/jenkins_home/workspace/Hiring-App1/target/hiring]
[0m[1;34mINFO@[0m] Processing war project
[0m[1;34mINFO@[0m] Copying webapp resources [/var/jenkins_home/workspace/Hiring-App1/src/main/webapp]
[0m[1;34mINFO@[0m] Building war: /var/jenkins_home/workspace/Hiring-App1/target/hiring.war
[0m[1;34mINFO@[0m] @[1m-----@[0m
[0m[1;34mINFO@[0m] @[1;32mBUILD SUCCESS@[0m
[0m[1;34mINFO@[0m] @[1m-----@[0m
[0m[1;34mINFO@[0m] Total time: 9.290 s
[0m[1;34mINFO@[0m] Finished at: 2026-05-20T07:49:12Z
[0m[1;34mINFO@[0m] @[1m-----@[0m
Finished: SUCCESS

• Nexus Artifactory

- Now create new EC2 for Nexus and connect it

☐	Karun	i-05952082T8516ab65	Stopped		t5.micro
☒	Nexus	i-0d66405dc536cf85b	Running		c7i-flex.large
☐	Docker	i-0be6ae4346131312d	Running		c7i-flex.large

- Installed java 1.8 version

```

[root@ip-172-31-19-102 ~]# yum install java-1.8.0-amazon-corretto -y
Last metadata expiration check: 0:54:10 ago on Wed May 20 07:53:52 2026.
Dependencies resolved.
=====
Package                Arch    Version                               Repository    Size
=====
Installing:
java-1.8.0-amazon-corretto  x86_64  1:1.8.0_492.b09-1.amzn2023          amazonlinux  38 M
Installing dependencies:
adwaita-cursor-theme      noarch  47-0.1.amzn2023.0.1                 amazonlinux  225 k

```

- Check java version

```
Enter to keep the current selection[+], or type selection number: 2
[root@ip-172-31-19-102 ~]# java -version
openjdk version "1.8.0_492"
OpenJDK Runtime Environment Corretto-8.492.09.1 (build 1.8.0_492-b09)
OpenJDK 64-Bit Server VM Corretto-8.492.09.1 (build 25.492-b09, mixed mode)
```

- Now Installed nexus

```
[root@ip-172-31-19-102 opt]# wget https://download.sonatype.com/nexus/3/nexus-3.70.1-02-unix.tar.gz
--2026-05-20 07:56:48-- https://download.sonatype.com/nexus/3/nexus-3.70.1-02-unix.tar.gz
Resolving download.sonatype.com (download.sonatype.com)... 184.169.209.110, 54.215.190.214
Connecting to download.sonatype.com (download.sonatype.com)|184.169.209.110|:443
... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.70.1-02-unix.tar.gz [following]
--2026-05-20 07:56:48-- https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.70.1-02-unix.tar.gz
Resolving cdn.download.sonatype.com (cdn.download.sonatype.com)... 104.18.1.134, 104.18.0.134, 2606:4700::6812:86, ...
Connecting to cdn.download.sonatype.com (cdn.download.sonatype.com)|104.18.1.134|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 256037928 (244M) [application/x-gzip]
Saving to: 'nexus-3.70.1-02-unix.tar.gz'

nexus-3.70.1-02-unix 100%[=====>] 244.18M 267MB/s in 0.9s

2026-05-20 07:56:49 (267 MB/s) - 'nexus-3.70.1-02-unix.tar.gz' saved [256037928/256037928]
```

- Now unzip or extract it

```
2026-05-20 07:56:49 (267 MB/s) - 'nexus-3.70.1-02-unix.tar.gz' saved [256037928/256037928]

[root@ip-172-31-19-102 opt]# ls
aws nexus-3.70.1-02-unix.tar.gz
[root@ip-172-31-19-102 opt]# tar -xvf nexus-3.70.1-02-unix.tar.gz
nexus-3.70.1-02/.install4j/9d17dc87.lprop
nexus-3.70.1-02/.install4j/MessagesDefault
nexus-3.70.1-02/.install4j/build.uuid
nexus-3.70.1-02/.install4j/e4ada6b7.lprop
nexus-3.70.1-02/.install4j/i4j_extf_0_17is1ik.utf8
```

- Rename it

```
ls
aws nexus-3.70.1-02 nexus-3.70.1-02-unix.tar.gz sonatype-work
[root@ip-172-31-19-102 opt]# mv nexus-3.70.1-02 nexus
[root@ip-172-31-19-102 opt]# ls
aws nexus nexus-3.70.1-02-unix.tar.gz sonatype-work
```

- create user add and password to it

```
, sharing)
[root@ip-172-31-19-102 ~]# useradd nexus
[root@ip-172-31-19-102 ~]# passwd nexus
Changing password for user nexus.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

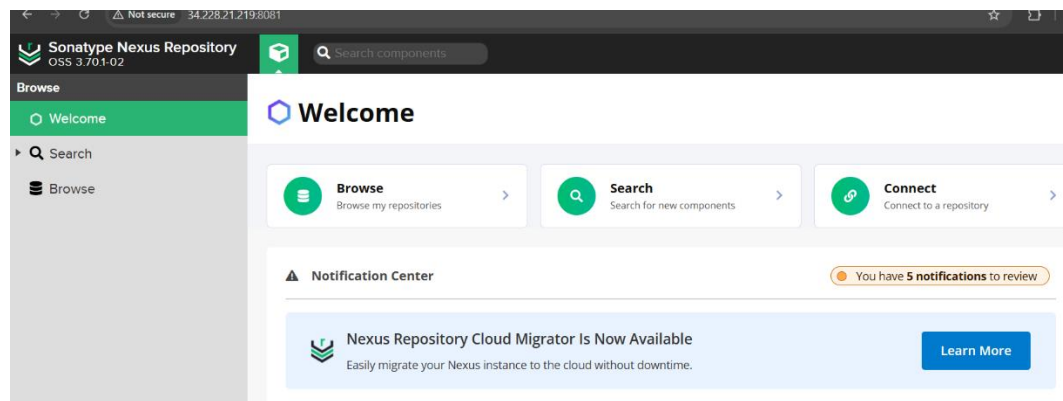
- Change permissions and login again to nexus

```
run_as_user="nexus"
[nexus@ip-172-31-19-102 nexus3]$ chown -R nexus:nexus /opt/sonatype-work
[nexus@ip-172-31-19-102 nexus3]$ chown -R nexus:nexus /opt/sonatype-work
[nexus@ip-172-31-19-102 nexus3]$ chmod -R 755 /opt/nexus
[nexus@ip-172-31-19-102 nexus3]$ su - nexus
Password:
Last login: Wed May 20 08:00:31 UTC 2026 on pts/1
```

- Start nexus and check running or not

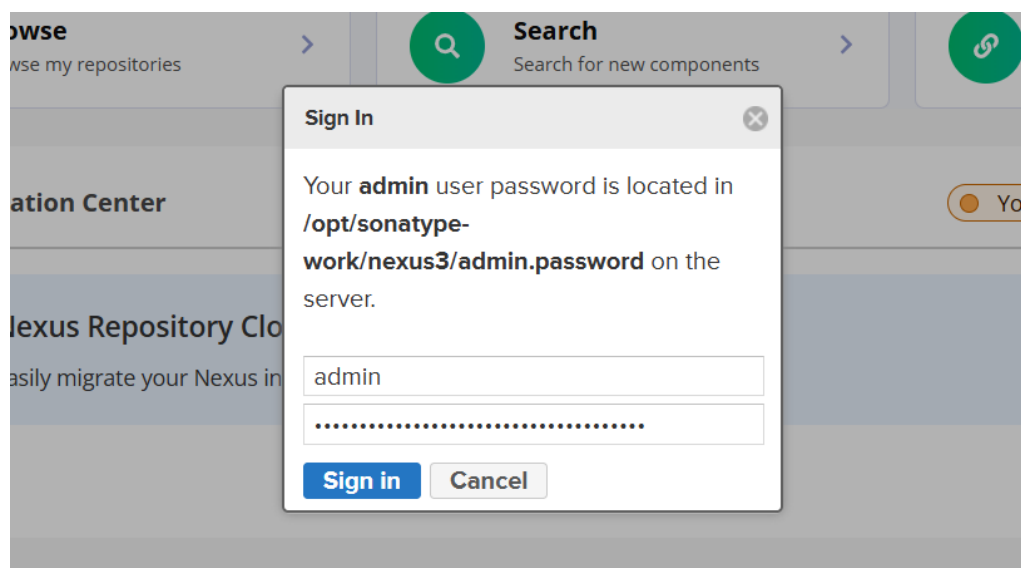
```
Last login: Wed May 20 08:45:14 UTC 2026 on pts/3
[nexus@ip-172-31-19-102 ~]$ /opt/nexus/bin/nexus start
Starting nexus
[nexus@ip-172-31-19-102 ~]$ /opt/nexus/bin/nexus status
nexus is running.
[nexus@ip-172-31-19-102 ~]$ /opt/nexus/bin/nexus status
nexus is running.
```

- Now with nexus ec2 public id and port number 8081 open in browser

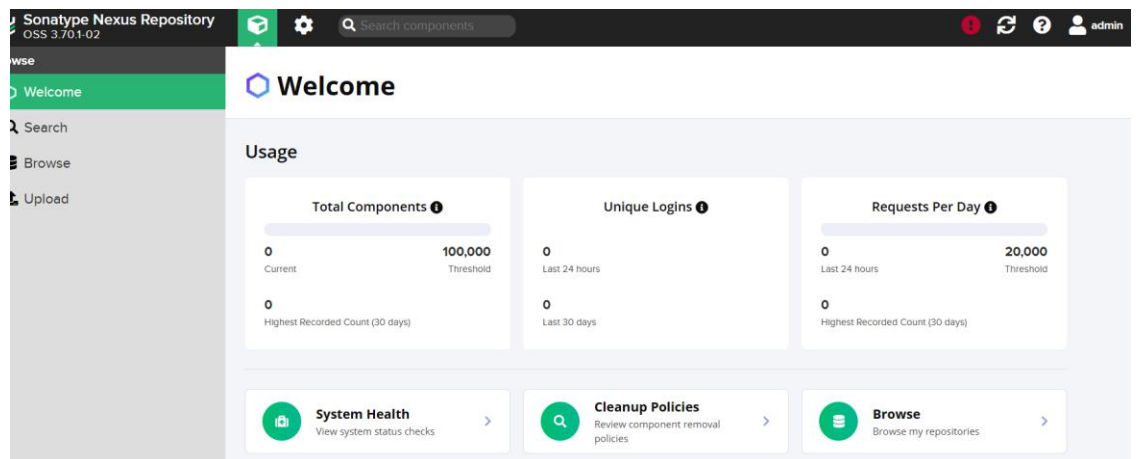


- generate a new password to it and login

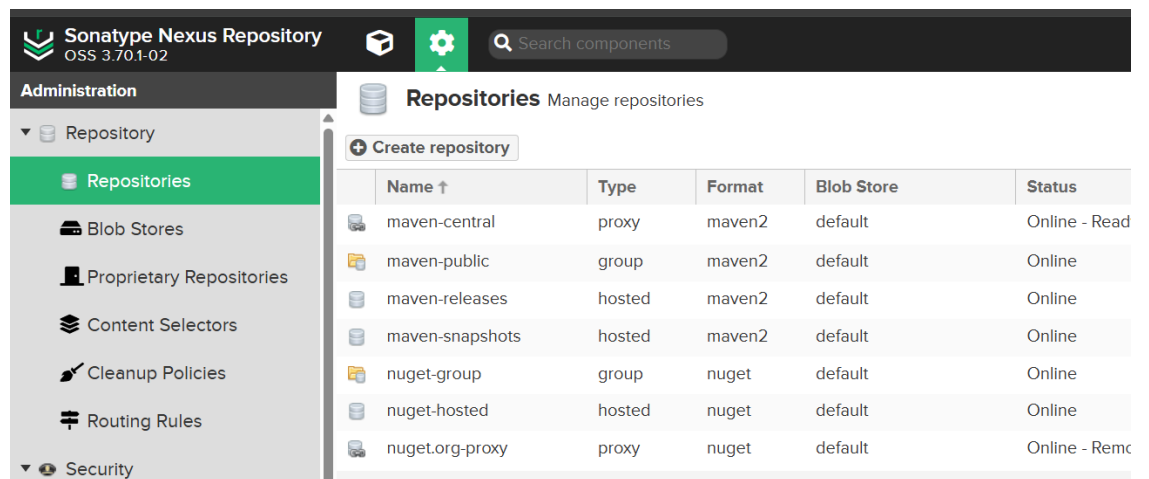
```
nexus is running.
[nexus@ip-172-31-19-102 ~]$ cat /opt/sonatype-work/nexus3/admin.password
7c5e92aa-53bf-4ef2-b978-0e3caa4a1b1c[nexus@ip-172-31-19-102 ~]$ client_loop
```



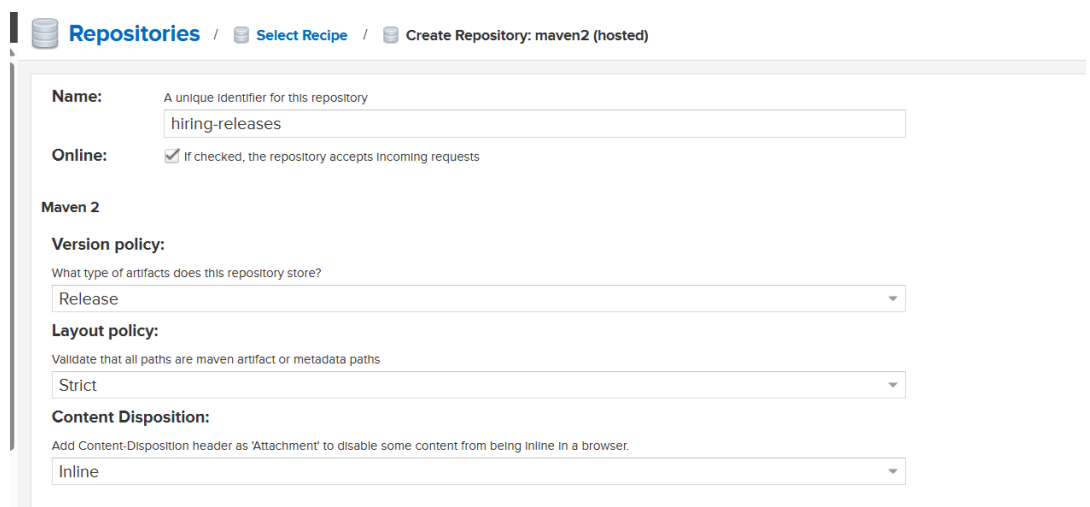
- Now create new password and save it and open home page



- Open repository and create new repository



- Given name as hiring-releases



- Set bob store as default and deployment policy as Allow redeploy

Storage

Blob store:
Blob store used to store repository contents
default

Strict Content Type Validation:
☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format

Hosted

Deployment policy:
Controls if deployments of and updates to artifacts are allowed
Allow redeploy

Proprietary Components:
☐ Components in this repository count as proprietary for namespace conflict attacks (requires Sonatype Nexus Firewall)

- Create a new repository successfully

Repository updated: hiring-releases

Name ↑	Type	Format	Blob Store	Status	URL	Health check	Firewall Re...
hiring-releases	hosted	maven2	default	Online			
maven-central	proxy	maven2	default	Online - Ready to Conn...		Analyze	
maven-public	group	maven2	default	Online			
maven-releases	hosted	maven2	default	Online			
maven-snapshots	hosted	maven2	default	Online			

- Open Jenkins in plugs install Nexus Artifactory uploader
- Open job and configure Build steps select nexus artifactory add details

nexus version
NEXUS3

Protocol
HTTP

Nexus URL ?
34.228.21.219:8081

Credentials
admin/***** + Add

GroupId
com.hiring

Version
1.0

- After giving all details save it and build

Repository ?

hiring-releases

Artifacts

Artifact

ArtifactId

hiring

Type ?

war

Classifier ?

File ?

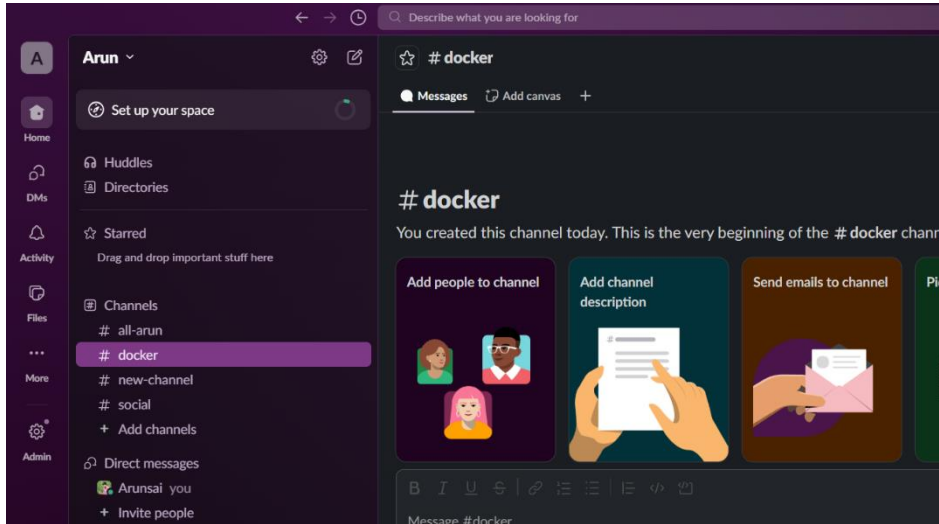
target/hiring.war

- After build check console as success

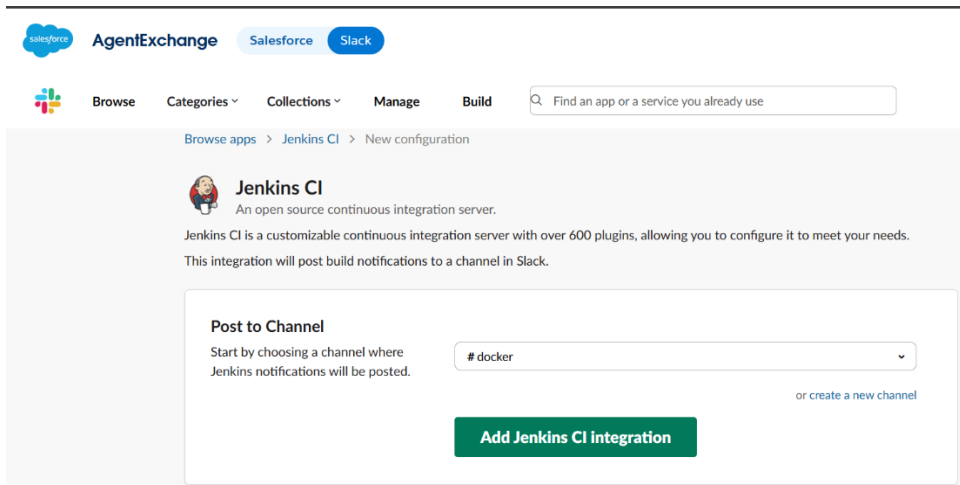
```
[1;34mINFO[m] [1m-----[m
Uploading artifact hiring.war started....
GroupId: com.hiring
ArtifactId: hiring
Classifier:
Type: war
Version: 1.0
File: hiring.war
Repository:hiring-releases
Uploading: http://34.228.21.219:8081/repository/hiring-releases/com/hiring/hiring/1.0/hiring-1.0.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://34.228.21.219:8081/repository/hiring-releases/com/hiring/hiring/1.0/hiring-1.0.war (1.7 kB at 7.6 kB/s)
Uploading artifact hiring.war completed.
Finished: SUCCESS
```

- **Slack Notification**

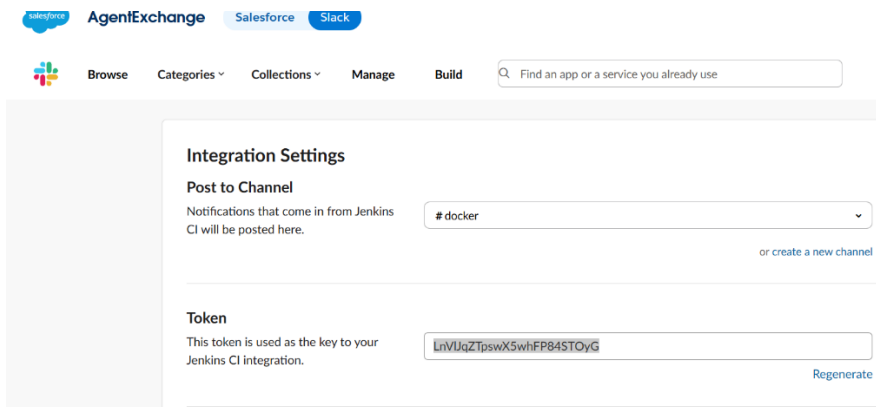
- Open new slack and create new workspace and channel



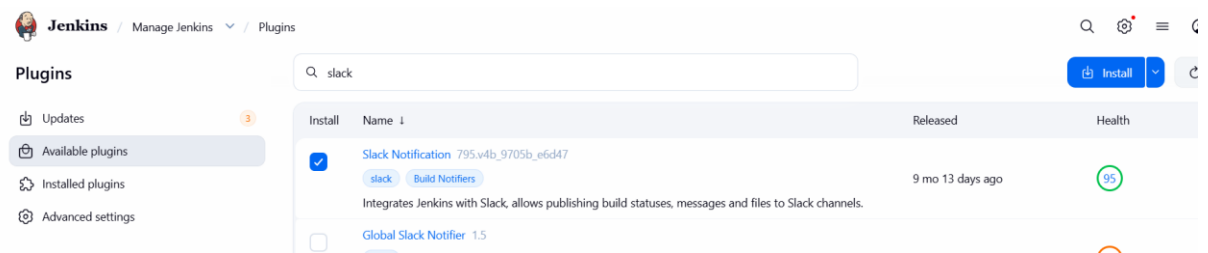
- In Apps install jenkins and given the channel name



- Generate token and with that token in jenkins create slack credentials



➤ Install slack notification in jenkins



➤ In manage jenkins system add details of slack

Slack

Workspace ?

arun-mmm8244

Credential ?

Slack

+ Add

Default channel / member id ?

C0B4YHQT24W

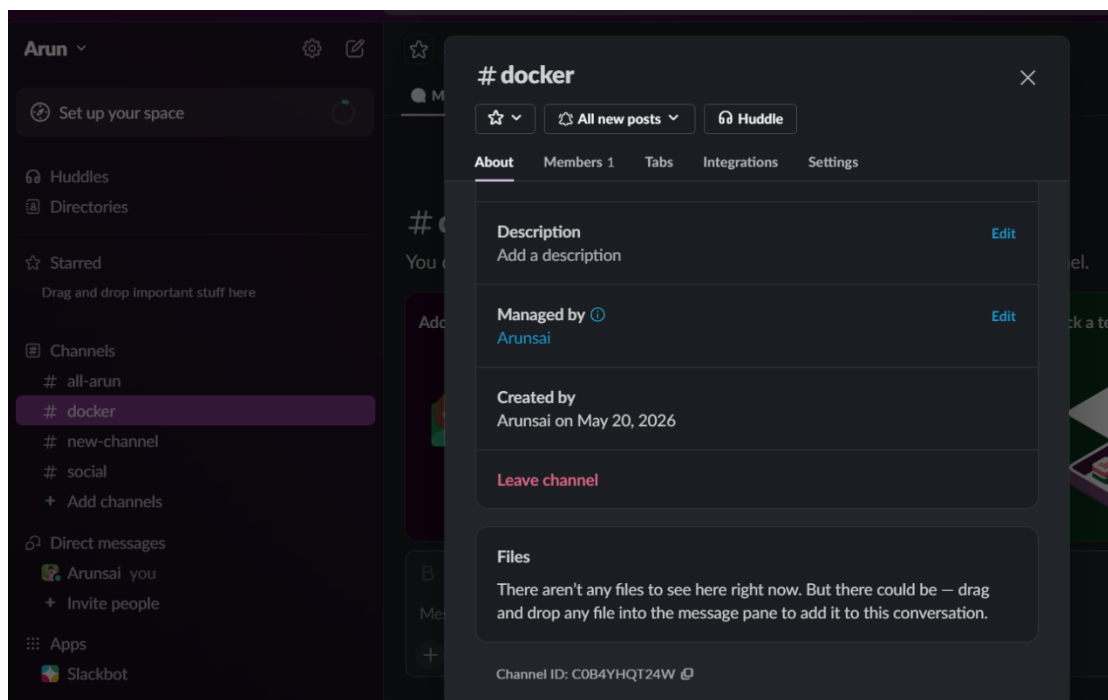
☐ Custom slack app bot user ?

Advanced ▾

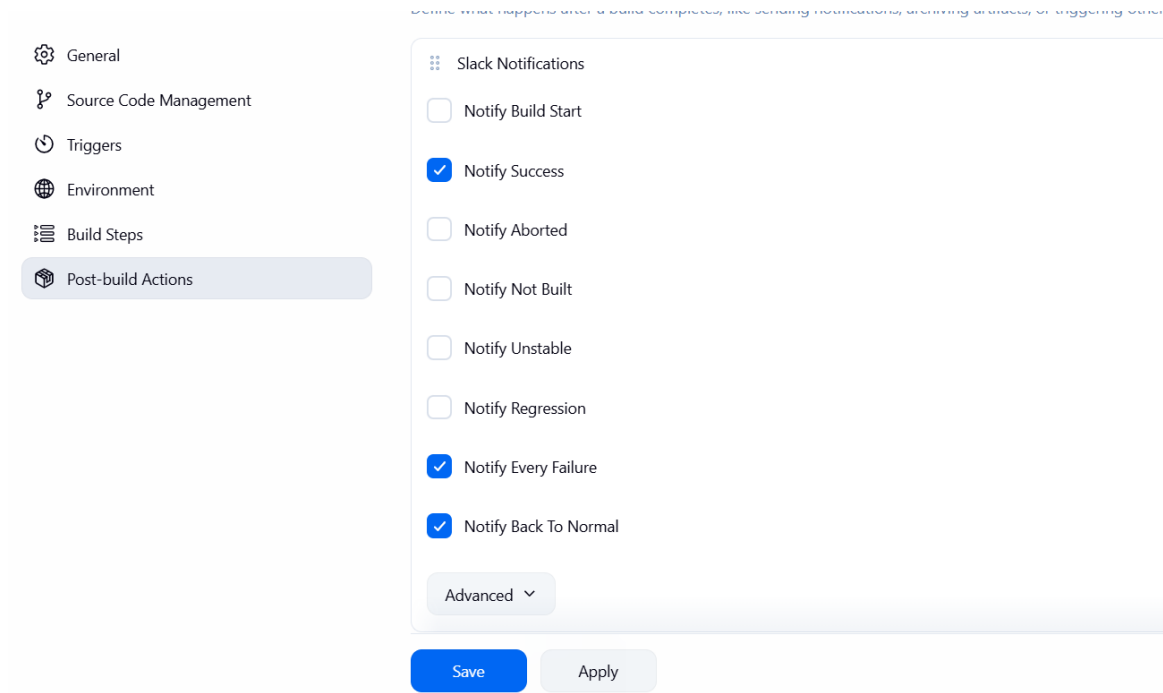
Success

Test Connection

➤ Channel member id below

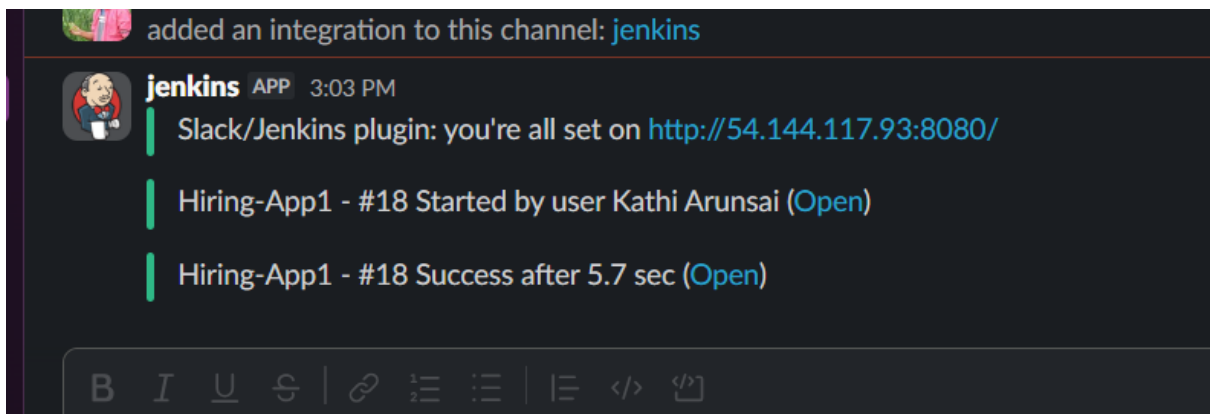


- Now in job do configure open post build action in slack notification enable this 3



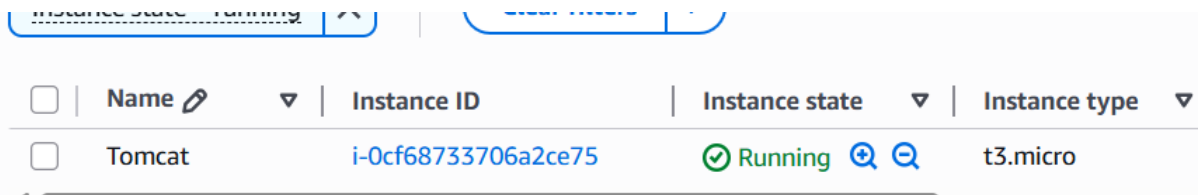
The screenshot shows the Jenkins configuration page for a job, specifically the 'Post-build Actions' tab. On the left, a sidebar lists various configuration options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions (which is highlighted). The main area displays the 'Slack Notifications' section. It contains several checkboxes for different build events: 'Notify Build Start' (unchecked), 'Notify Success' (checked), 'Notify Aborted' (unchecked), 'Notify Not Built' (unchecked), 'Notify Unstable' (unchecked), 'Notify Regression' (unchecked), 'Notify Every Failure' (checked), and 'Notify Back To Normal' (checked). Below these checkboxes is an 'Advanced' dropdown menu. At the bottom of the configuration area are two buttons: 'Save' and 'Apply'.

- Then we get slack notification in docker channel when Success or Failure in Build



- **Deploy On tomcat**

- Create a new ec2 instance for tomcat and connect it



The screenshot shows a table from the AWS Management Console displaying details for an EC2 instance. The table has columns for Name, Instance ID, Instance state, and Instance type. There is one instance listed with the name 'Tomcat'.

☐	Name	Instance ID	Instance state	Instance type
☐	Tomcat	i-0cf68733706a2ce75	Running	t3.micro

- Installed Java 17 in tomcat

```
[ec2-user@ip-172-31-17-23 ~]$ sudo su -
[root@ip-172-31-17-23 ~]# yum install java-17-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 311 kB/s | 38 kB 00:00
Last metadata expiration check: 0:00:01 ago on Wed May 20 09:39:48 2026.
Dependencies resolved.
=====
Package                        Arch  Version                                Repository  Size
=====
Installing:
  java-17-amazon-corretto      x86_64 1:17.0.19+10-1.amzn2023.1 amazonlinux 219 k
Installing dependencies:
```

- Then after install tomcat through link

```
[root@ip-172-31-17-23 ~]# cd /opt
[root@ip-172-31-17-23 opt]# wget https://archive.apache.org/dist/tomcat/tomcat-9
/v9.0.98/bin/apache-tomcat-9.0.98.tar.gz
--2026-05-20 09:40:56-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.98
/bin/apache-tomcat-9.0.98.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a
:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... con
nected.
HTTP request sent, awaiting response... 200 OK
Length: 12760610 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.98.tar.gz'

apache-tomcat-9.0.9 100%[=====>] 12.17M 177KB/s in 1m 50s

2026-05-20 09:42:47 (113 KB/s) - 'apache-tomcat-9.0.98.tar.gz' saved [12760610/1
2760610]

[root@ip-172-31-17-23 opt]# tar -xvf apache-tomcat-9.0.98.tar.gz
apache-tomcat-9.0.98/conf/
apache-tomcat-9.0.98/conf/catalina.policy
apache-tomcat-9.0.98/conf/catalina.properties
apache-tomcat-9.0.98/conf/context.xml
apache-tomcat-9.0.98/conf/jaspic-providers.xml
apache-tomcat-9.0.98/conf/jaspic-providers.xsd
apache-tomcat-9.0.98/conf/logging.properties
```

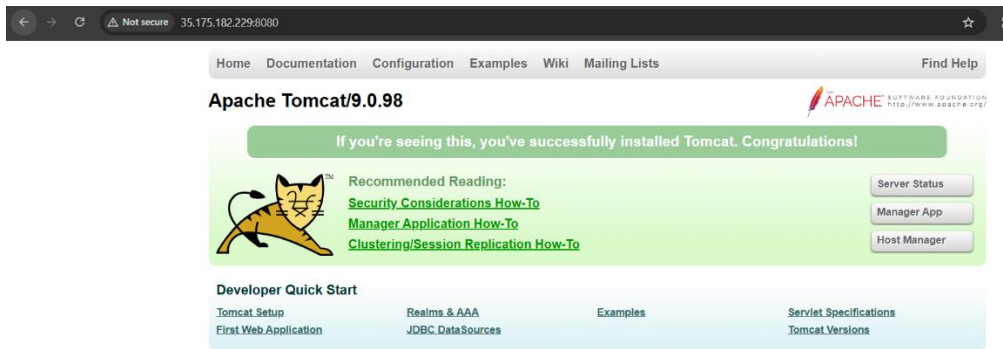
- Then after that unzip the tomcat

```
[root@ip-172-31-17-23 opt]# tar -xvf apache-tomcat-9.0.98.tar.gz
apache-tomcat-9.0.98/conf/
apache-tomcat-9.0.98/conf/catalina.policy
apache-tomcat-9.0.98/conf/catalina.properties
apache-tomcat-9.0.98/conf/context.xml
apache-tomcat-9.0.98/conf/jaspic-providers.xml
apache-tomcat-9.0.98/conf/jaspic-providers.xsd
apache-tomcat-9.0.98/conf/logging.properties
```

- Rename the tomcat and check port no

```
[root@ip-172-31-17-23 opt]# mv apache-tomcat-9.0.98 tomcat
[root@ip-172-31-17-23 opt]# /opt/tomcat/bin/startup.sh
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli
.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-17-23 opt]# ps -ef | grep tomcat
root      25740      1   39   09:43 pts/1    00:00:03 /usr/bin/java -Djava.util.lo
gging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manage
r=org.apache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava
.protocol.handler.pkgs=org.apache.catalina.webresources -Dorg.apache.catalina.se
curity.SecurityListener.UMASK=0027 -Dignore.endorsed.dirs= -classpath /opt/tomca
t/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/opt/tomcat
-Dcatalina.home=/opt/tomcat -Djava.io.tmpdir=/opt/tomcat/temp org.apache.catalin
a.startup.Bootstrap start
```

- Now start tomcat and check status and open in browser



- Open manager and host manager file and change as comment the lines at below

```
[root@ip-172-31-17-23 opt]# vim /opt/tomcat/webapps/manager/META-INF/context.xml
[root@ip-172-31-17-23 opt]# vim /opt/tomcat/webapps/host-manager/META-INF/context.xml
[root@ip-172-31-17-23 opt]# /opt/tomcat/bin/shutdown.sh
/opt/tomcat/bin/startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED
```

- Change to Comment the value and allow lines

```

        sameSiteCookies="strict" />
<!--
<Valve className="org.apache.catalina.valves.RemoteAddrValve"
  allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />
-->
<Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|string)|org\.apache\.catalina\.filters\.CsrfPreventionFilter|he(?:\.$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>

```

- Open tomcat-users and add username and password lines into it.

```

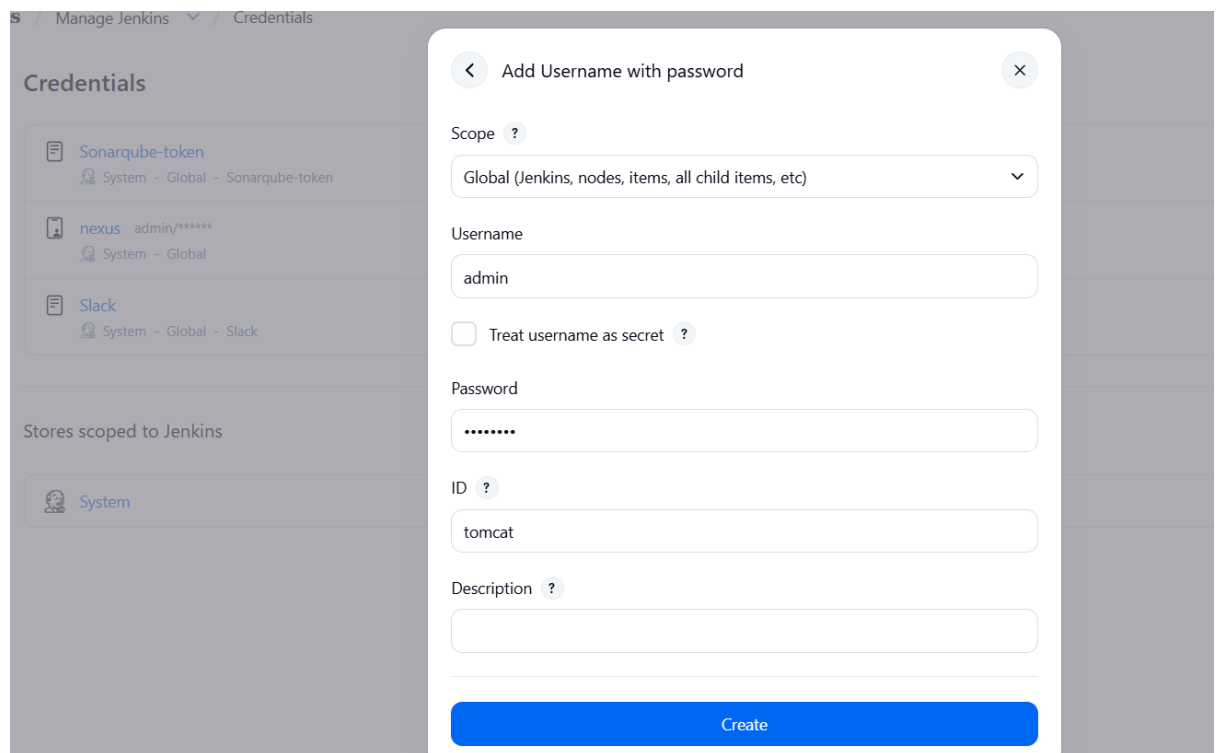
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
  version="1.0">

  <role rolename="manager-gui"/>
  <role rolename="manager-script"/>
  <user username="tomcat" password="tomcat123" roles="manager-gui,manager-script"/>

<!--
By default, no user is included in the "manager-gui" role required
to operate the "/manager/html" web application.  If you wish to use this app

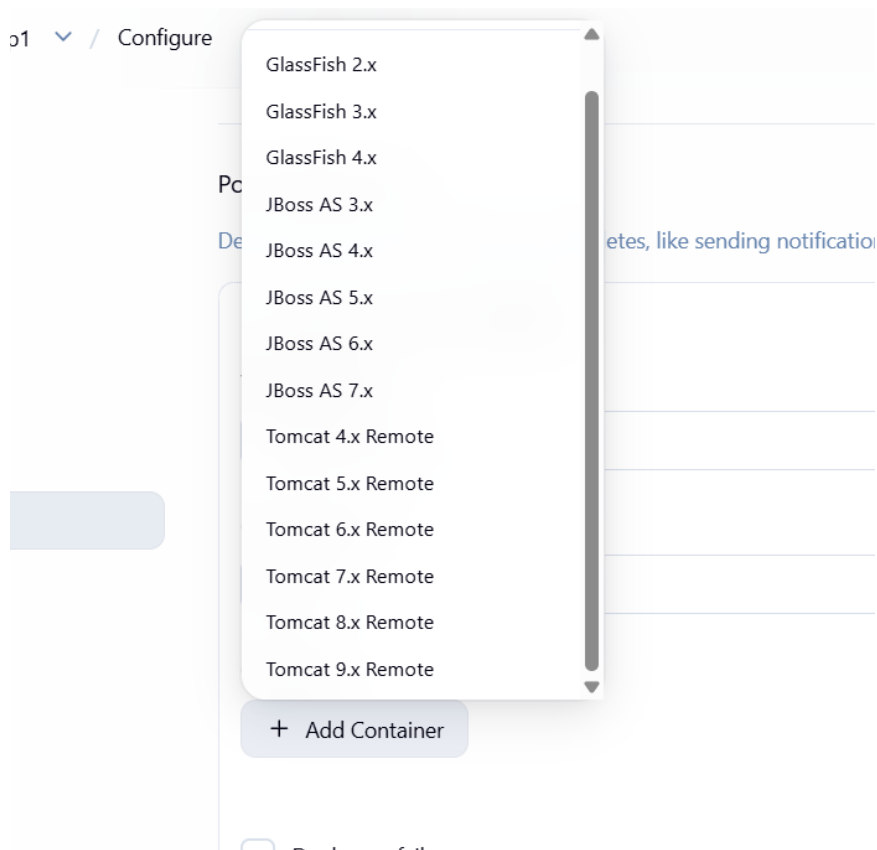
```

- Now open jenkins and create token for tomcat credentials



The screenshot shows the Jenkins 'Credentials' configuration page. On the left, there's a sidebar with 'Manage Jenkins' and 'Credentials' tabs. Below, a list of credentials is shown: 'Sonarqube-token' (System - Global - Sonarqube-token), 'nexus' (admin/*, System - Global), and 'Slack' (System - Global - Slack). Below this, 'Stores scoped to Jenkins' shows 'System'. The main area is a modal titled 'Add Username with password'. It has fields for 'Scope' (Global (Jenkins, nodes, items, all child items, etc)), 'Username' (admin), 'Password' (masked with dots), 'ID' (tomcat), and 'Description'. There's a checkbox 'Treat username as secret' which is unchecked. A blue 'Create' button is at the bottom.

- Now plugins install deploy war/ear plugin
- Open job and configure post build actions and add these below details tomcat 9.x



The screenshot shows the Jenkins job configuration page, specifically the 'Post-build Actions' section. A dropdown menu is open, showing a list of container images: GlassFish 2.x, GlassFish 3.x, GlassFish 4.x, JBoss AS 3.x, JBoss AS 4.x, JBoss AS 5.x, JBoss AS 6.x, JBoss AS 7.x, Tomcat 4.x Remote, Tomcat 5.x Remote, Tomcat 6.x Remote, Tomcat 7.x Remote, Tomcat 8.x Remote, and Tomcat 9.x Remote. Below the list is a button '+ Add Container'. The background shows the 'Configure' page with a 'Post-build Actions' section and a 'Deploy' section.

➤ Deploy war/ear function

Deploy war/ear to a container

WAR/EAR files ?

target/hiring.war

Context path ?

hiring

Containers

Tomcat 9.x Remote

Credentials

Tomcat/*****

Tomcat URL ?

http://35.175.182.229:8080

➤ Build now and check console

Jenkins / Hiring-App1 / #22

```
[01;34mINFO[m] [1m-----[m]
[01;34mINFO[m] Total time: 1.422 s
[01;34mINFO[m] Finished at: 2026-05-20T10:26:53Z
[01;34mINFO[m] [1m-----[m]
Uploading artifact hiring.war started....
GroupId: com.hiring
ArtifactId: hiring
Classifier:
Type: war
Version: 1.0
File: hiring.war
Repository:hiring-releases
Uploading: http://34.228.21.219:8081/repository/hiring-releases/com/hiring/hiring/1.0/hiring-1.0.war
100 % completed (1.7 kB / 1.7 kB).
uploaded: http://34.228.21.219:8081/repository/hiring-releases/com/hiring/hiring/1.0/hiring-1.0.war (1.7 kB at 25 kB/s)
Uploading artifact hiring.war completed.
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/jenkins_home/workspace/Hiring-App1/target/hiring.war to container Tomcat 9.x Remote with context hiring
[/var/jenkins_home/workspace/Hiring-App1/target/hiring.war] is not deployed. Doing a fresh deployment.
Deploying [/var/jenkins_home/workspace/Hiring-App1/target/hiring.war]
[Slack Notifications] found #21 as previous completed, non-aborted build
[Slack Notifications] will send OnBackToNormalNotification because build matches and user preferences allow it
Finished: SUCCESS
```

➤ After success open manager/html page and login the page

← → ↻ Not secure 35.175.182.229:8080/manager/html



Tomcat Web Application Manager

Message: OK

Manager

List ApplicationsHTML Manager HelpManager HelpServer Status

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

- And Finally Run the URL Job and check it and done with 1 task



2) Setup a Jenkins CICD pipeline using Declarative pipeline using feature-1.1 branch.

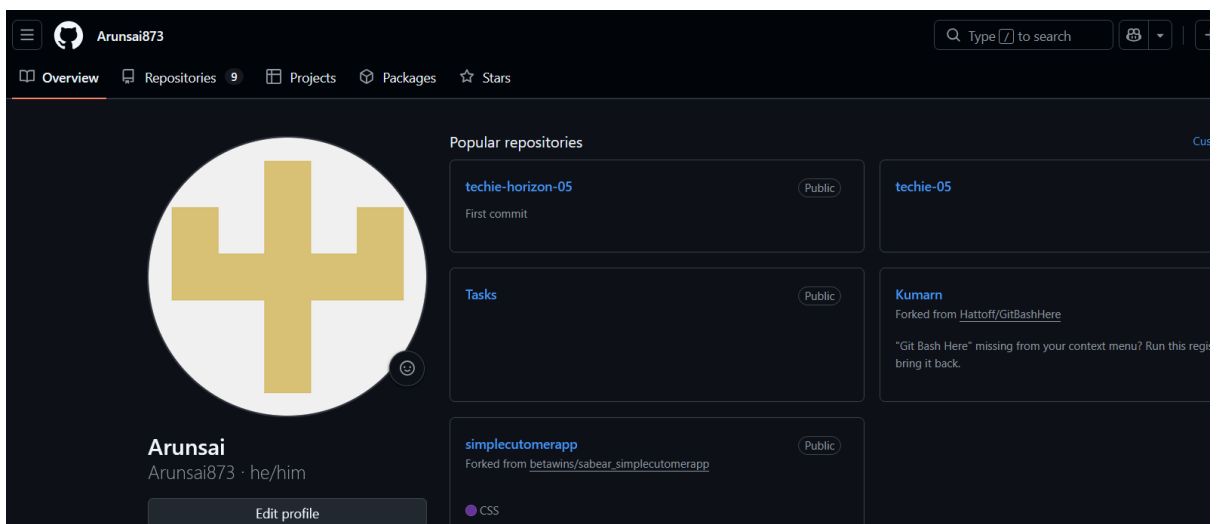
Source Code:

https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1

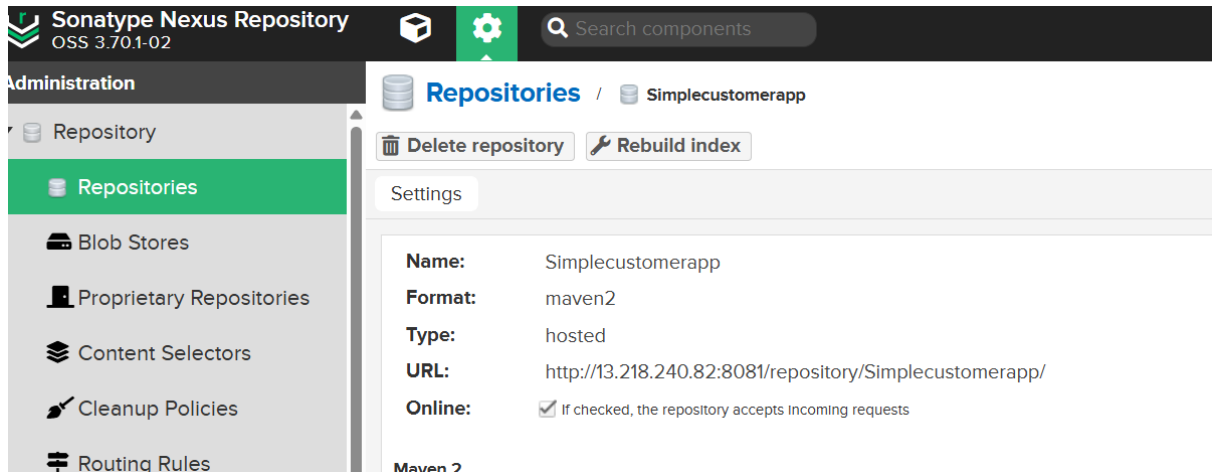
stages:

- Git Clone
- SonarQube Integration
- Maven Compilation
- Nexus Artifactory
- Slack Notification
- Deploy On tomcat

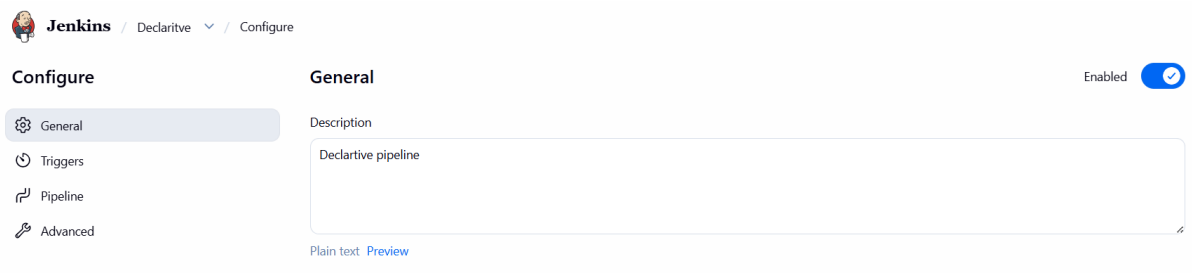
- First fork the source code into my git repo



- In Nexus created a new repo as Simplecustomerapp



- Already SonarQube, sonar-scanner, maven, nexus, tomcat all created and done with plugins and configurations
- Create a New Item and selected pipeline



- Now written the declarative pipeline and save it and build now

```
pipeline {
  agent any
  tools {
    maven 'Maven'
  }
  stages {
    stage('Clean Workspace') {
      steps {
        cleanWs()
      }
    }
    stage('Clone Code') {
      steps {
        git branch: 'master',
        url: 'https://github.com/Arunsai873/simplecutomerapp.git'
      }
    }
  }
}
```

```

stage('Build using Maven') {
    steps {
        sh 'mvn clean install'
    }
}
stage('Publish to Nexus') {
    steps {
        script {
            def pom = readMavenPom file: 'pom.xml'
            def releaseVersion = pom.version.replace("-SNAPSHOT", "")
            def artifactPath = "target/${pom.artifactId}-${pom.version}.war"
            echo "Uploading Artifact: ${artifactPath}"
            nexusArtifactUploader(
                nexusVersion: 'nexus3',
                protocol: 'http',
                nexusUrl: '13.218.240.82:8081',
                repository: 'Simplecustomerapp',
                groupId: pom.groupId,
                version: releaseVersion,
                credentialsId: 'nexus',
                artifacts: [
                    [
                        artifactId: pom.artifactId,
                        classifier: "",
                        file: artifactPath,
                        type: 'war'
                    ]
                ]
            )
        }
    }
}
stage('Deploy On Tomcat') {
    steps {
        script {
            sh 'ls target/*.war | head -1 > warfile.txt'
            def warFile = readFile('warfile.txt').trim()
            echo "Deploying WAR: ${warFile}"
            deploy adapters: [
                tomcat9(
                    credentialsId: 'tomcat',
                    path: "",
                    url: 'http://54.204.140.94:8080/manager/text'
                )
            ]
        }
    }
}

```

```

    ],
    contextPath: 'arunsai',
    war: warFile
  }
}
}
}
}
post {
  success {
    slackSend(
      tokenCredentialId: 'Slack',
      channel: 'C0B5EQ639SQ',
      color: 'good',
      message: "SUCCESS: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"
    )
  }
  failure {
    slackSend(
      tokenCredentialId: 'Slack',
      channel: 'C0B5EQ639SQ',
      color: 'danger',
      message: "FAILED: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"
    )
  }
}
}
}

```

➤ pipeline success

```
[Pipeline] slackSend
```

```
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: arun-mmm8244, channel: C0B5EQ639SQ, color: good, botUser: false, tokenCredentialId: Slack, notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
```

```
[Pipeline] }
```

```
[Pipeline] // stage
```

```
[Pipeline] }
```

```
[Pipeline] // withEnv
```

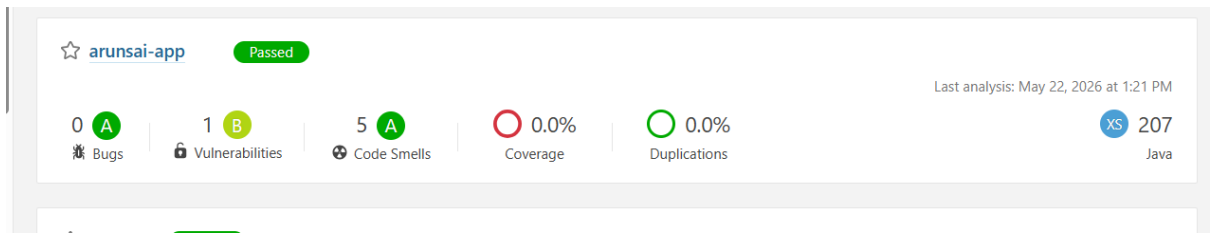
```
[Pipeline] }
```

```
[Pipeline] // node
```

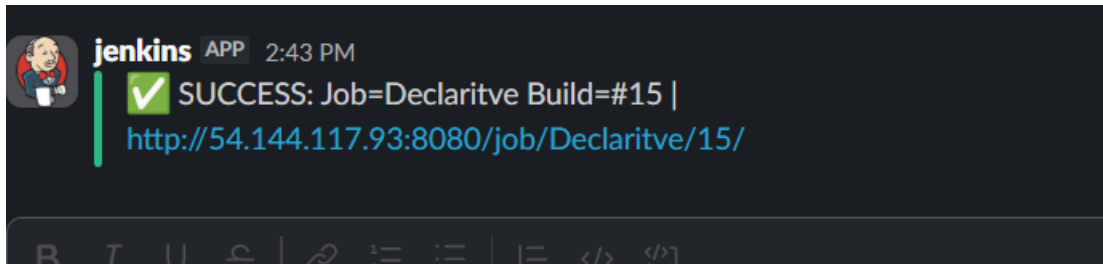
```
[Pipeline] End of Pipeline
```

```
Finished: SUCCESS
```

- We can check in SonarQube properties



- We can check in slack for notification



3) Setup a Jenkins CI/CD pipeline using Scripted pipeline using feature-1.1 branch.

Source Code:

https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1

stages:

- **Git Clone**
 - **SonarQube Integration**
 - **Maven Compilation**
 - **Nexus Artifactory**
 - **Slack Notification**
 - **Deploy On tomcat**
- Created a new item for scripted and selected pipeline

The Jenkins 'New Item' configuration page is shown. The 'Enter an item name' field contains 'Scripted'. Under 'Select an item type', the 'Pipeline' option is selected. The other options are:

- Freestyle project**: Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**: Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project**: Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

- Already fork and created repo in nexus now written scripted pipeline and save it

```
node {  
  
    def mvnHome = tool 'Maven'  
  
    env.PATH = "${mvnHome}/bin:${env.PATH}"  
  
    try {  
  
        stage('Clean Workspace') {  
  
            cleanWs()  
  
        }  
  
        stage('Clone Code') {  
  
            git branch: 'master',  
  
            url: 'https://github.com/Arunsai873/simplecutomerapp.git'  
  
        }  
  
        stage('Build using Maven') {  
  
            sh 'mvn clean install'  
  
        }  
  
        stage('Publish to Nexus') {  
  
            def pom = readMavenPom file: 'pom.xml'  
  
            def releaseVersion = pom.version.replace("-SNAPSHOT", "")  
  
            def artifactPath = "target/${pom.artifactId}-${pom.version}.war"  
  
            echo "Uploading Artifact: ${artifactPath}"  
  
        }  
    }  
}
```

```

nexusArtifactUploader(
    nexusVersion: 'nexus3',
    protocol: 'http',
    nexusUrl: '13.218.240.82:8081',
    repository: 'Simplecustomerapp',
    groupId: pom.groupId,
    version: releaseVersion,
    credentialsId: 'nexus',
    artifacts: [
        [
            artifactId: pom.artifactId,
            classifier: "",
            file: artifactPath,
            type: 'war'
        ]
    ]
)
}

```

```

stage('Deploy On Tomcat') {
    sh 'ls target/*.war | head -1 > warfile.txt'

    def warFile = readFile('warfile.txt').trim()

    echo "Deploying WAR: ${warFile}"

    deploy adapters: [
        tomcat9(

```

```
        credentialsId: 'tomcat',

        path: '',

        url: 'http://54.204.140.94:8080/manager/text'

    )

},

contextPath: 'arunsai',

war: warFile

}
```

```
// Success Slack Notification
```

```
slackSend(

    tokenCredentialId: 'Slack',

    channel: 'C0B5EQ639SQ',

    color: 'good',

    message: "SUCCESS: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"

)
```

```
} catch (Exception e) {
```

```
// Failure Slack Notification
```

```
slackSend(

    tokenCredentialId: 'Slack',

    channel: 'C0B5EQ639SQ',

    color: 'danger',

    message: "FAILED: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"

)
```

```

    )

    throw e // rethrow so Jenkins marks build as FAILED

}

}

```

- After written script build and check console

```

[Pipeline] // stage
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: arun-mmm8244, channel: C0B5EQ639SQ, color: good, botUser: false,
tokenCredentialId: Slack, notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

- Now check in slack for notification



4) Write sample skeleton of pipelines.

- Declarative Pipeline

```

pipeline {
  agent any
  tools {
    maven 'Maven'
  }
  stages {
    stage('Clean Workspace') {
      steps {
        cleanWs()
      }
    }
    stage('Clone Code') {
      steps {
        git branch: 'master',
        url: 'https://github.com/Arunsai873/simplecutomerapp.git'
      }
    }
  }
}

```

```

}
stage('Build using Maven') {
    steps {
        sh 'mvn clean install'
    }
}
stage('Publish to Nexus') {
    steps {
        script {
            def pom = readMavenPom file: 'pom.xml'
            def releaseVersion = pom.version.replace("-SNAPSHOT", "")
            def artifactPath = "target/${pom.artifactId}-${pom.version}.war"
            echo "Uploading Artifact: ${artifactPath}"
            nexusArtifactUploader(
                nexusVersion: 'nexus3',
                protocol: 'http',
                nexusUrl: '13.218.240.82:8081',
                repository: 'Simplecustomerapp',
                groupId: pom.groupId,
                version: releaseVersion,
                credentialsId: 'nexus',
                artifacts: [
                    [
                        artifactId: pom.artifactId,
                        classifier: "",
                        file: artifactPath,
                        type: 'war'
                    ]
                ]
            )
        }
    }
}
stage('Deploy On Tomcat') {
    steps {
        script {
            sh 'ls target/*.war | head -1 > warfile.txt'
            def warFile = readFile('warfile.txt').trim()
            echo "Deploying WAR: ${warFile}"
            deploy adapters: [
                tomcat9(
                    credentialsId: 'tomcat',
                    path: "",
                    url: 'http://54.204.140.94:8080/manager/text'
                )
            ]
        }
    }
}

```

```

        )
    ],
    contextPath: 'arunsai',
    war: warFile
}
}
}
}
post {
    success {
        slackSend(
            tokenCredentialId: 'Slack',
            channel: 'C0B5EQ639SQ',
            color: 'good',
            message: "SUCCESS: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"
        )
    }
    failure {
        slackSend(
            tokenCredentialId: 'Slack',
            channel: 'C0B5EQ639SQ',
            color: 'danger',
            message: "FAILED: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"
        )
    }
}
}
}

```

➤ Scripted Pipeline

```

node {

    def mvnHome = tool 'Maven'

    env.PATH = "${mvnHome}/bin:${env.PATH}"

    try {

        stage('Clean Workspace') {

```

```
    cleanWs()
}

stage('Clone Code') {
    git branch: 'master',
    url: 'https://github.com/Arunsai873/simplecutomerapp.git'
}

stage('Build using Maven') {
    sh 'mvn clean install'
}

stage('Publish to Nexus') {
    def pom = readMavenPom file: 'pom.xml'
    def releaseVersion = pom.version.replace("-SNAPSHOT", "")
    def artifactPath = "target/${pom.artifactId}-${pom.version}.war"
    echo "Uploading Artifact: ${artifactPath}"
    nexusArtifactUploader(
        nexusVersion: 'nexus3',
        protocol: 'http',
        nexusUrl: '13.218.240.82:8081',
        repository: 'Simplecustomerapp',
        groupId: pom.groupId,
        version: releaseVersion,
        credentialsId: 'nexus',
```

```
artifacts: [  
  [  
    artifactId: pom.artifactId,  
    classifier: "",  
    file: artifactPath,  
    type: 'war'  
  ]  
]  
)  
}
```

```
stage('Deploy On Tomcat') {  
  sh 'ls target/*.war | head -1 > warfile.txt'  
  def warFile = readFile('warfile.txt').trim()  
  echo "Deploying WAR: ${warFile}"  
  deploy adapters: [  
    tomcat9(  
      credentialsId: 'tomcat',  
      path: "",  
      url: 'http://54.204.140.94:8080/manager/text'  
    )  
  ],  
  contextPath: 'arunsai',  
  war: warFile  
}
```

```

// Success Slack Notification

slackSend(

    tokenCredentialId: 'Slack',

    channel: 'C0B5EQ639SQ',

    color: 'good',

    message: "SUCCESS: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"

)

} catch (Exception e) {

// Failure Slack Notification

slackSend(

    tokenCredentialId: 'Slack',

    channel: 'C0B5EQ639SQ',

    color: 'danger',

    message: "FAILED: Job=${env.JOB_NAME}
Build=#${env.BUILD_NUMBER} | ${env.BUILD_URL}"

)

    throw e // rethrow so Jenkins marks build as FAILED

}

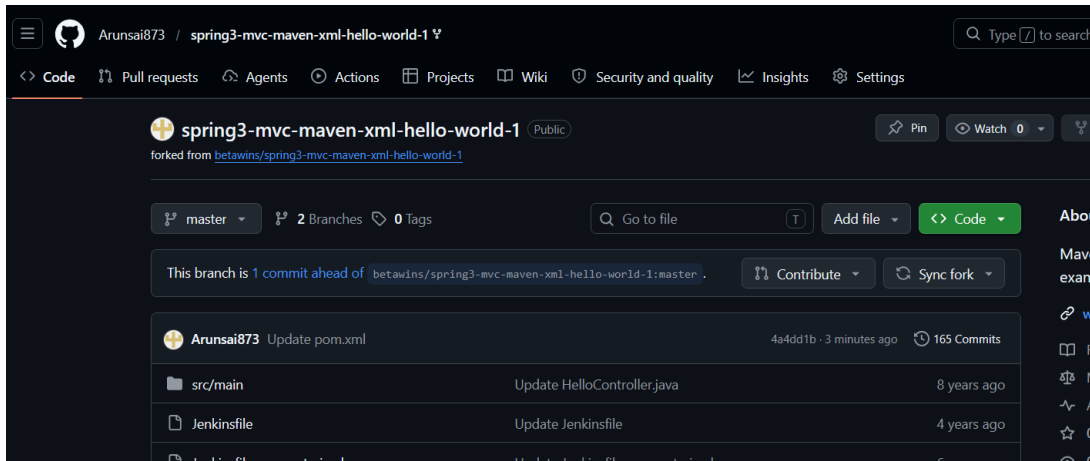
}

```

5) Create a parameterized job in Jenkins.

Source Code: <https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git>

- First fork the git into my repo



- Created a new item in Jenkins and selected freestyle project

New Item

Enter an item name

Parameterized-SpringMVC

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

- Selected Parameter and add parameter



Jenkins

Parameterized-SpringMVC

Configure

Configure

General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

General

Description

Parameterized-SpringMVC

Plain text Preview

☐ Discard old builds ?

☐ GitHub project

☒ This project is parameterised ?

+ Add Parameter

- In parameter selected string parameter added branch and master

Save the project

☒ This project is parameterised ?

String Parameter ?

Name ?

BRANCH

Default Value ?

master

Definition ?

- In Source code management select git and add URL

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/Arunsai873/spring3-mvc-maven-xml-hello-world-1.git

Credentials ?

- In Build steps select maven and given maven version and save

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Invoke top-level Maven targets ?

Maven Version

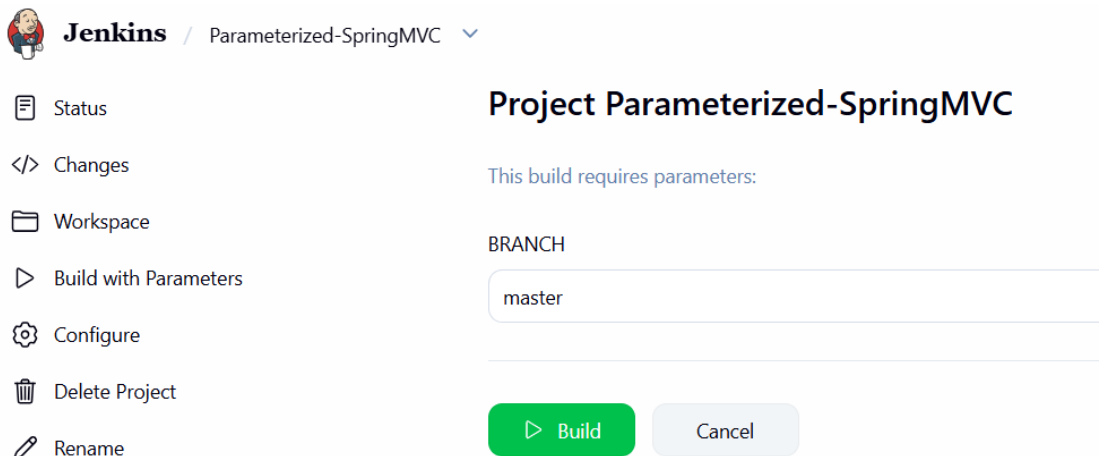
Maven

Goals

clean package

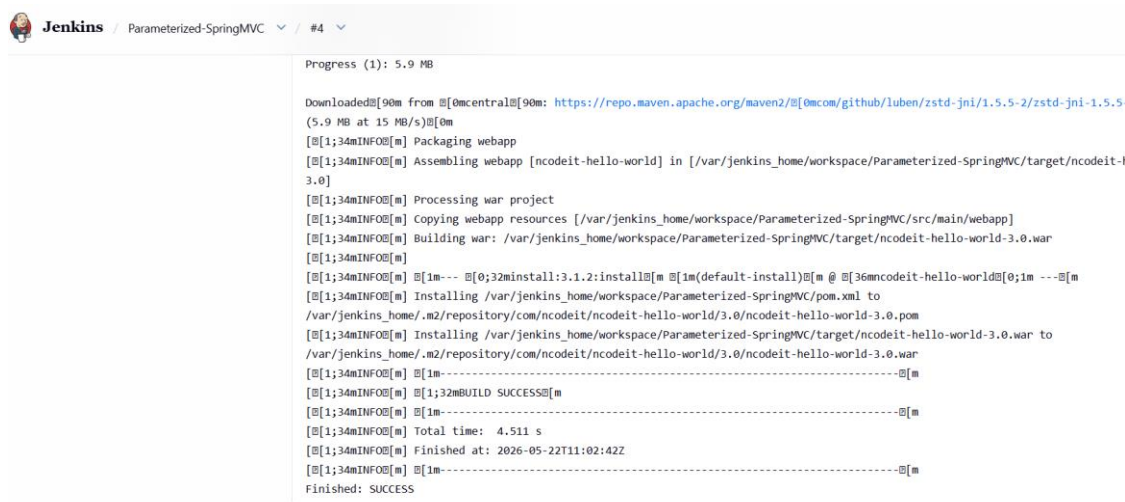
Advanced ▾

➤ After save Build with parameters



The screenshot shows the Jenkins web interface for a project named "Parameterized-SpringMVC". On the left, there is a sidebar with navigation links: Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, and Rename. The main area displays the project name "Project Parameterized-SpringMVC" and a message "This build requires parameters:". Below this, there is a "BRANCH" field with the value "master". At the bottom right, there are two buttons: "Build" (green) and "Cancel" (grey).

➤ In console we can check Build success

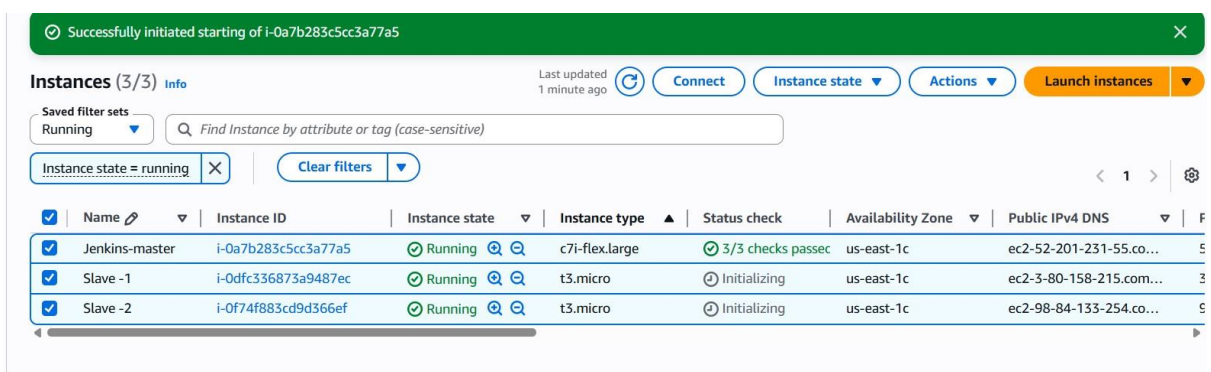


The screenshot shows the Jenkins console output for a build. The output text is as follows:

```
Progress (1): 5.9 MB
Downloaded[90m from @0mcentral@90m: https://repo.maven.apache.org/maven2/@0mcom/github/luben/zstd-jni/1.5.5-2/zstd-jni-1.5.5-
(5.9 MB at 15 MB/s)@0m
[01;34mINFO@0m] Packaging webapp
[01;34mINFO@0m] Assembling webapp [ncodeit-hello-world] in [/var/jenkins_home/workspace/Parameterized-SpringMVC/target/ncodeit-h
3.0]
[01;34mINFO@0m] Processing war project
[01;34mINFO@0m] Copying webapp resources [/var/jenkins_home/workspace/Parameterized-SpringMVC/src/main/webapp]
[01;34mINFO@0m] Building war: /var/jenkins_home/workspace/Parameterized-SpringMVC/target/ncodeit-hello-world-3.0.war
[01;34mINFO@0m]
[01;34mINFO@0m] @1m--- @0;32minstall:3.1.2:install@0m @1m(default-install)@0m @ @36mncodeit-hello-world@0;1m ---@0m
[01;34mINFO@0m] Installing /var/jenkins_home/workspace/Parameterized-SpringMVC/pom.xml to
/var/jenkins_home/.m2/repository/com/ncodeit/ncodeit-hello-world/3.0/ncodeit-hello-world-3.0.pom
[01;34mINFO@0m] Installing /var/jenkins_home/workspace/Parameterized-SpringMVC/target/ncodeit-hello-world-3.0.war to
/var/jenkins_home/.m2/repository/com/ncodeit/ncodeit-hello-world/3.0/ncodeit-hello-world-3.0.war
[01;34mINFO@0m] @1m-----@0m
[01;34mINFO@0m] @1;32mBUILD SUCCESS@0m
[01;34mINFO@0m] @1m-----@0m
[01;34mINFO@0m] Total time: 4.511 s
[01;34mINFO@0m] Finished at: 2026-05-22T11:02:42Z
[01;34mINFO@0m] @1m-----@0m
Finished: SUCCESS
```

6) Setup one slave machine for Jenkins.

➤ Created 2EC2 instances for 2 slaves and for master used old one



The screenshot shows the AWS Management Console "Instances" page. At the top, there is a green notification bar: "Successfully initiated starting of i-0a7b283c5cc3a77a5". Below this, the "Instances (3/3)" section shows a table of three EC2 instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, and Public IPv4 DNS. The instances are:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Jenkins-master	i-0a7b283c5cc3a77a5	Running	c7i-flex.large	3/3 checks passed	us-east-1c	ec2-52-201-231-55.co...
Slave -1	i-0dfc336873a9487ec	Running	t3.micro	Initializing	us-east-1c	ec2-3-80-158-215.com...
Slave -2	i-0f74f883cd9d366ef	Running	t3.micro	Initializing	us-east-1c	ec2-98-84-133-254.co...

- In slave 1 installed java 21

```
[ec2-user@ip-172-31-26-202 ~]$ sudo yum install java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 319 kB/s | 38 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1.amzn2023.0.3 amazonlinux 503 k
```

- In slave 2 installed java 21

```
[ec2-user@ip-172-31-27-165 ~]$ sudo yum install java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 320 kB/s | 38 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1.amzn2023.0.3 amazonlinux 503 k
```

- Now in Jenkins master in root added user add and passwd Jenkins

```
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-21-129 ~]$ sudo su -
[root@ip-172-31-21-129 ~]# sudo passwd jenkins
Changing password for user jenkins.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-21-129 ~]# id -u jenkins &>/dev/null || useradd -m -s /bin/bash -d /var/
lib/jenkins jenkins
[root@ip-172-31-21-129 ~]# sudo passwd jenkins
Changing password for user jenkins.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

- In master created ssh keygen in su – Jenkins

```
[ec2-user@ip-172-31-26-202 ~]$ sudo su - jenkins
Last login: Wed May 13 15:54:43 UTC 2026 on pts/2
[jenkins@ip-172-31-26-202 ~]$ mkdir -p ~/.ssh
[jenkins@ip-172-31-26-202 ~]$ ssh-keyscan -H 54.89.9.228 >> ~/.ssh/known_hosts
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
[jenkins@ip-172-31-26-202 ~]$ chmod 700 ~/.ssh
[jenkins@ip-172-31-26-202 ~]$ chmod 600 ~/.ssh/known_hosts
[jenkins@ip-172-31-26-202 ~]$ cat ~/.ssh/known_hosts
|1|A2ExNKe/k1QRLDp9fhqJbCocf7o=|95N1gN4qPR2Gx6dFCynw7TWefXs= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBERYUfoscwVc
WoudbnObwAomUuZ7I4Xoxj8HJg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|mYumLDrudZMynykrQg89tqvZ0kA=|AT0xZj7IqY7dvrJh+dDt6kVnnNU= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbXhA8GZX35J7ZTSASouiqws6j5S+iUBVw1eNwWdJN
|1|KHuOgd/tyPrPSoqW9c1aMz+61js=|ZwHGIxo+xOsfCb0vtupeZwpfcr4= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBERYUfoscwVc
WoudbnObwAomUuZ7I4Xoxj8HJg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|TR479ACwDUxpnrcteaBf8/OqLU=|7eTeUM1LEgSuxJT3dAejD1g21xM= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbXhA8GZX35J7ZTSASouiqws6j5S+iUBVw1eNwWdJN
[jenkins@ip-172-31-26-202 ~]$ exit
logout
```

- In Slave 1 added the Jenkin master key given permissions

```
mbNhhHwYmVtP+Yy1BqNP+mm+IE2NnYXknq/fcNn5HFJgDK+HgKtq5KcvtraTQ==" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-26-202 ~]# mkdir -p /home/jenkins/.ssh
[root@ip-172-31-26-202 ~]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQC2ub0yBCTPndFN7AhoI7Sjg9fMY9jCjVzn5f4SoMMaA
VERNZ47HEeNAbOIFLkywjmVXcwTjOk47tIf25xXC7AwVORn+dZ1dYowVdcf1pjf0i01Up3vtI7H8eL8R2ScT76S0GmXgbGyyXk1eJJuQ3LXrk/xEWP
XRCpE6ui3l0Y3s0CiJCbPnAQIknxpvZ+OwrS1o7fjHM12NhS6NhAmwr/Nfhwrr+v5Ud2Ja0g6g6ebGcQ2P5eQqoA/wEZegmQMPBILD5tACKgTFJqDm
VbyJEATCyPDPpWBk++tfuLrNVyQU8f8ntNgmRDK/V/QsQAtjNjSOYnArAhmfEwtuuzkc0g1/fvo1NLZNcAS/+gB4t9fU5V1YrG8/E0kU6uzYTutsXm+
Tpb1fLyC9VxsDVTdDnyS0uvWaNKN1nrzGmaZlUPaIKY69EwrsTl016GtrqUNf+pn2z7zSxwh2cg7qQbP8N5TmrVusTBe4HsZ40N+N2Z8bg/+l0IP1sY
e1r+gBfwIdpIr7FqLU+WnNpc1n5I/ZNDrhe1hXohUE1mPBL8M+Usk+SQwVZsf9M71GjNCJtm+S613kwtZTUWJSDWzztNA1xj+bz30PiSwIdFOF907A
KwBwR2qh+mBNBnhHwyMvtP+Yy1BqNP+mm+IE2NnYXknq/fcNn5HFJgDK+HgKtq5KcvtraTQ==" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-26-202 ~]# chown jenkins:jenkins /home/jenkins/.ssh/authorized_keys
chown: invalid user: 'jenkins:jenkins'
[root@ip-172-31-26-202 ~]# chown jenkins:jenkins /home/jenkins/.ssh
chown: invalid user: 'jenkins:jenkins'
[root@ip-172-31-26-202 ~]# useradd -m -s /bin/bash jenkins
useradd: warning: the home directory /home/jenkins already exists.
useradd: Not copying any file from skel directory into it.
[root@ip-172-31-26-202 ~]# chown jenkins:jenkins /home/jenkins/.ssh
[root@ip-172-31-26-202 ~]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-26-202 ~]# exit
logout
```

- Same in Slave 2 added the Jenkin master key and given permissions

```
[ec2-user@ip-172-31-27-165 ~]$ sudo su -
[root@ip-172-31-27-165 ~]# mkdir -p /home/jenkins/.ssh
[root@ip-172-31-27-165 ~]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQC2ub0yBCTPndFN7AhoI7Sjg9fMY9jCjVzn5f4SoMMaAVERNZ47HEeNAbOIFLkywjmVXcwTjOk47tIf25xXC7AwVORn+dZ1dYowVdcf1pjf0i01U
p3vtI7H8eL8R2ScT76S0GmXgbGyyXk1eJJuQ3LXrk/xEWPXRCpE6ui3l0Y3s0CiJCbPnAQIknxpvZ+OwrS1o7fjHM12NhS6NhAmwr/Nfhwrr+v5Ud2Ja0g6g6ebGcQ2P5eQqoA/wEZegmQMPBILD5tACKgTFJqDmVbyJEATCyPDPpWBk++tfuLr
NVyQU8f8ntNgmRDK/V/QsQAtjNjSOYnArAhmfEwtuuzkc0g1/fvo1NLZNcAS/+gB4t9fU5V1YrG8/E0kU6uzYTutsXm+Tpb1fLyC9VxsDVTdDnyS0uvWaNKN1nrzGmaZlUPaIKY69EwrsTl016GtrqUNf+pn2z7zSxwh2cg7qQbP8N5TmrVusTB
e4HsZ40N+N2Z8bg/+l0IP1sYe1r+gBfwIdpIr7FqLU+WnNpc1n5I/ZNDrhe1hXohUE1mPBL8M+Usk+SQwVZsf9M71GjNCJtm+S613kwtZTUWJSDWzztNA1xj+bz30PiSwIdFOF907AKwBwR2qh+mBNBnhHwyMvtP+Yy1BqNP+mm+IE2NnYXknq
/fcNn5HFJgDK+HgKtq5KcvtraTQ==" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-27-165 ~]# useradd -m -s /bin/bash jenkins
useradd: warning: the home directory /home/jenkins already exists.
useradd: Not copying any file from skel directory into it.
[root@ip-172-31-27-165 ~]# chown jenkins:jenkins /home/jenkins/.ssh
[root@ip-172-31-27-165 ~]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-27-165 ~]# exit
logout
```

- In master connect with ssh-key scan of slave1 public id known hosts

```
[ec2-user@ip-172-31-21-129 ~]$ sudo su - jenkins
Last login: Wed May 13 15:14:39 UTC 2026 on pts/1
[jenkins@ip-172-31-21-129 ~]$ mkdir -p ~/.ssh
[jenkins@ip-172-31-21-129 ~]$ ssh-keyscan -H 54.89.9.228 >> ~/.ssh/known_hosts
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
[jenkins@ip-172-31-21-129 ~]$ chmod 700 ~/.ssh
[jenkins@ip-172-31-21-129 ~]$ chmod 600 ~/.ssh/known_hosts
[jenkins@ip-172-31-21-129 ~]$ cat ~/.ssh/known_hosts
[1]YyuroBK00tI1od8oGSGqt/BnqmQ=|A+cAh9xnhXGQBDnZrzCv0+8vc40= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlkZHAyNTYAAAAIbmlkZHAyNTYAAABBBERYUfoscwVo
woudbn0bwaOmUuz7I4XoxjBHJg5zr6saQn3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98k56e/saGUU=
[1]ErApp4kz2BkQCrAf0T08XyRImM=|3Cbn1+wjgGrd7vzwF4RuMsnot40= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbXhA8GZx35J7zTSASouiqws6j5S+iUBVw1eNwvdJN
[1]at9p4TB+B0f9ELzerJjjoYxdgFs=|3bCkfInLfw/do/z/6ZKVJkgXxI= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI0kIpJnwJ2U1oZD/t6qDNKuMHYvqc/1MTysd2E3R7ort
[1]TZkxtHYousErw/e3u4I7Z55UijM=|X8HeQUACf7lZDsA4NoTsCMdm07E= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlkZHAyNTYAAAAIbmlkZHAyNTYAAABBBJA7/O/DXreU
koOMLzhKIEXns9a3dUuaLKmpDFX28uyOCJ4VBdFO0UDIYyZvhrYAKm/9RXemtRGagGisPW93dg=
[1]R6QC1mkuU9/FLJP0dsR0hb1GN4=|Zgz6Niki65oeSzoUvgBmx9GmXVs= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI0kIpJnwJ2U1oZD/t6qDNKuMHYvqc/1MTysd2E3R7ort
[1]b1T/371TjTtOUMURYjyJtBSYT0R8=|+0bnpk2rhM2wc4mEgFMvsOVkLew= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlkZHAyNTYAAAAIbmlkZHAyNTYAAABBBJA7/O/DXreU
koOMLzhKIEXns9a3dUuaLKmpDFX28uyOCJ4VBdFO0UDIYyZvhrYAKm/9RXemtRGagGisPW93dg=
[1]av426f1xs1N5QDMQ/6MCZaHBLgE=|rS8VdWmqKpTeTpCv3zYEZVnuU= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbXhA8GZx35J7zTSASouiqws6j5S+iUBVw1eNwvdJN
[1]aFJempVYXU0gx+nunyXFG+nDdIw=|T5fw8ay1cPrSm54zETHmsGKyI94= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlkZHAyNTYAAAAIbmlkZHAyNTYAAABBBERYUfoscwVo
woudbn0bwaOmUuz7I4XoxjBHJg5zr6saQn3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98k56e/saGUU=
```

- Now open Jenkins master and create new node for slave1

 Jenkins

Nodes

Slave-1

Configure

Delete Agent

Configure

Build History

Load Statistics

Log

Open Blue Ocean

Build Executor Status

(0 of 1 executor busy)

Name

Slave-1

Description

Slave-1 Machine

Plain text Preview

Number of executors

1

Remote root directory

/home/ec2-user/jenkins/

- Given label, host slave 1 public and create credentials with ec2 pem

Labels ?

linux slave1 docker

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

54.89.9.228

Credentials ?

ec2-user (Slave1) + Add

Host Key Verification Strategy ?

Known hosts file Verification Strategy

- Save it and launch or relaunch agent check the status

Jenkins / Nodes / Slave-1

Status

Delete Agent

Configure

Build History

Load Statistics

Script Console

Log

System Information

Disconnect

Open Blue Ocean

Agent Slave-1

Slave-1 Machine

Disk space is below threshold of 1.00 GiB. Only 458.37 MiB out of 458.40 MiB left on /tmp.

Monitoring Data

Labels

linux slave1 docker

Projects tied to Slave-1

None

- Check logs and agent successfully connected

Jenkins / Nodes / Slave-1 / Log

```
{
  { alias;
    eval ${which_declare} ) | /usr/bin/which --tty-only --read-alias --read-functions --show-tilde --show-dot "$@"
  }
}
[05/13/26 16:03:06] [SSH] Starting sftp client.
[05/13/26 16:03:06] [SSH] Copying latest remoting.jar...
Source agent hash is 7A9A77A64F02EB9F50BCEFF9E9A349E2. Installed agent hash is 7A9A77A64F02EB9F50BCEFF9E9A349E2
Verified agent jar. No update is necessary.
Expanded the channel window size to 4MB
[05/13/26 16:03:06] [SSH] Starting agent process: cd "/home/ec2-user/jenkins" && java -jar remoting.jar -workDir /home/ec2-user/jenkins -jar-cache /home/ec2-user/jenkins/remoting/jarCache
May 13, 2026 4:03:06 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ec2-user/jenkins/remoting as a remoting work directory
May 13, 2026 4:03:06 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ec2-user/jenkins/remoting
<====[JENKINS REMOTING CAPACITY]====>channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: standard in/out
This is a Unix agent
Agent successfully connected and online
```

- Now same with Slave 2 In master connect with ssh-key scan of slave 2 private id known hosts (first logout from all roots)

```
[ec2-user@ip-172-31-21-129 ~]$ sudo su - jenkins
Last login: Wed May 13 12:13:47 UTC 2026 on pts/2
[jenkins@ip-172-31-21-129 ~]$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
[jenkins@ip-172-31-21-129 ~]$ ssh-keyscan -H 172.31.27.165 >> ~/.ssh/known_hosts
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
[jenkins@ip-172-31-21-129 ~]$ chmod 600 ~/.ssh/known_hosts
[jenkins@ip-172-31-21-129 ~]$ cat ~/.ssh/known_hosts
|1|VYuroBKO0tiIod8oGSGgt/BnqmQ=|A+cAh9xnHxGQ8dnZrzcv0iBvc40= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlldHAYNTYAAAAIbmlldHAYNTYAAABBBERYUfoscwVo
WoudbnObwAomUUZ7I4XoxjBHJg5Zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|ErApp4kBz28kqCrAfQTO8XyRlNm=|3CbnI+wjgGrd/vzwF4RuMsnot4Q= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHmbhxA8GZX35J7zTSASouiqws6j5S+iUBVwleNwWdJN
|1|at9p4TB+B0F9ELzerJJjoYxdgFs=|JbCkFInLFwb/do/z/6ZKVJkgXxI= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOkIpJnwJ2U10zD/t6qDNKuMHYvqc/1MTysd2E3R7ort
|1|TZkxtHyOusErw7e3U4I7Z55UijM=|X8HeQUACF71ZDsA4NoTsCMdm07E= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlldHAYNTYAAAAIbmlldHAYNTYAAABBBJA7/0/DXreu
KoOMLzhKIExns9a3DUualKmpDFX28uyOCJ4VBdDF00UDIyyZvhrYAKm/9RXemtRGaqGisPW93dg=
```

- Now open Jenkins master and create new node for slave2

jenkins / Nodes ▾ / Slave-2 ▾ / Configure

is

te Agent

figure

History

Statistics

n Blue Ocean

utor Status

utor busy)

Name ?

Slave-2

Description ?

Slave-2 Machine

Plain text [Preview](#)

Number of executors ?

1

Remote root directory ?

/home/ec2-user/jenkins

- Given label, host slave 2 private id and create credentials with ec2 pem (Same pem key of slave 1)

Labels ?

linux docker

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

172.31.27.165

Credentials ?

ec2-user (Slave2) + Add

Host Key Verification Strategy ?

Known hosts file Verification Strategy

➤ Save it and launch the bring this node check the status

Jenkins / Nodes ▾ / Slave-2

status

Delete Agent

Configure

Build History

Load Statistics

Log

Open Blue Ocean

Executor Status

executor busy)

Agent Slave-2

Slave-2 Machine

! Disk space is below threshold of 1.00 GiB. Only 458.24 MiB out of 458.40 MiB left on /tmp.

Monitoring Data ▾

Labels

linux docker

Projects tied to Slave-2

None

Edit description Bring this node back online

➤ Check the logs Successfully connected

Jenkins / Nodes ▾ / Slave-2 ▾ / Log

```
which ()
{
    ( alias;
      eval ${which_declare} ) | /usr/bin/which --tty-only --read-alias --read-functions --show-tilde --show-dot "$@"
    }
[05/13/26 15:25:48] [SSH] Starting sftp client.
[05/13/26 15:25:48] [SSH] Copying latest remoting.jar...
[05/13/26 15:25:48] [SSH] Copied 1,407,936 bytes.
Expanded the channel window size to 4MB
[05/13/26 15:25:48] [SSH] Starting agent process: cd "/home/ec2-user/jenkins" && java -jar remoting.jar -workDir /home/ec2-user/jenkins -j
cache /home/ec2-user/jenkins/remoting/jarCache
May 13, 2026 3:25:48 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ec2-user/jenkins/remoting as a remoting work directory
May 13, 2026 3:25:49 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ec2-user/jenkins/remoting
<===[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```